

Imam Mohammad ibn Saud Islamic University

College of Computer and Information Sciences

Information Systems Department

Awan-أوان

By:

Shahad Asem Alnasser,

And 2 other students participate in this project

Supervised by:

Alaa

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Abstract

Cooperative training plays a vital role in enabling students to get practical experience and to be ready for the job market. But the cooperative training process is accompanied by a lot of problems, such as the inability to find training opportunities and poor communication among students as well as the training organization and the university supervisor. Moreover, there are difficulties in handling applications, supervision, and the coordination of training activities effectively. The requirement gathering techniques and issues mentioned show the need for a more organized and centralized solution that will enhance the overall cooperative training experience.

This project presents "Awan," a smart cooperative training platform designed to connect students, universities, supervisors, and companies through one centralized system. The platform aims to simplify and improve the cooperative training journey, starting from searching and applying for opportunities to communication, evaluations, progress monitoring, and report submission. The proposed system also provides AI-based recommendations to help students discover more suitable opportunities efficiently.

To better understand the current cooperative training process and stakeholder needs, several requirement gathering techniques were used, including background analysis, questionnaires, and interviews. The collected results provided insights into the current workflow, common challenges, stakeholder expectations, and suggested improvements related to communication, opportunity management, and student follow-up.

Based on these findings, the proposed Awan system was designed to support the entire cooperative training process through features such as opportunity posting and approval, application management, notifications, messaging, progress report submission, progress monitoring, and student evaluation, helping create a more organized, efficient, and connected training experience.

الملخص

يقوم التدريب التعاوني بدور هام لمساعدة الطلاب على اكتساب خبرة عملية والاستعداد لسوق العمل مع ذلك يواجه العديد من الطلاب صعوبات خلال عملية التدريب التعاوني، كإيجاد فرص مناسبة وضعف التواصل ومحدودية المتابعة طوال فترة التدريب إضافةً إلى ذلك، غالبًا ما تواجه مؤسسات التدريب والمشرفون الجامعيون تحديات في إدارة الطلبات ومتابعة تقدم الطلاب وتنسيق أنشطة التدريب بكفاءة توضح هذه المشكلات الحاجة إلى حل أكثر تنظيمًا ومركزية، ليحسن تجربة التدريب التعاوني بشكل عام .يقدم مشروع (أوان) منصة ذكية للتدريب التعاوني مصممة لربط الطلاب والجامعات والمشرفين والشركات من خلال نظام مركزي واحد وتهدف المنصة إلى تبسيط وتحسين رحلة التدريب التعاوني يبدأ من البحث عن الفرص والتقدم إليها مرورا بالتواصل والتقييمات ومتابعة التقدم وصولًا إلى تقديم التقارير كما يوفر النظام المقترح توصيات قائمة على الذكاء الاصطناعي لمساعدة الطلاب على اكتشاف فرص أكثر ملاءمة بكفاءة .لتحسين فهم عملية التدريب التعاوني الحالية واحتياجات الجهات المعنية تم استخدام العديد من التقنيات لجمع المتطلبات بما في ذلك تحليل الأنظمة الموجودة والاستبيانات والمقابلات، قدمت النتائج التي تم جمعها نظره ثاقبة حول سير العمل الحالي والتحديات الشائعة وتوقعات الجهات المعنية واقتراحات لتحسينات تتعلق بالتواصل، وإدارة الفرص، ومتابعة الطلاب .وبناءً على هذه النتائج، صُمم نظام (أوان) المقترح لدعم عملية التدريب التعاوني بأكملها من خلال ميزات مثل نشر الفرص والموافقة عليها وإدارة الطلبات والإشعارات، والمراسلة، وتقديم تقارير التقدم، ومراقبة التقدم، وتقييم الطلاب مما يُسهم في خلق تجربة تدريبية أكثر تنظيمًا وكفاءة وترابطًا ..

Keywords

Cooperative Training
Training Opportunities
Awan System
Awan Website
Students
Company Supervisors
University Supervisors
University Training Unit
Companies
Organizations
AI-based Recommendations
Performance Monitoring
Stakeholder Communication
Application Management
Opportunity Management
Progress Tracking
Requirement Gathering
Background Analysis
Intelligent Matching
Notifications
Report Submission
Training Organizations
Web Application
Internship Management System

List of Abbreviations

AI : Artificial Intelligence

BPMN : Business Process Model and Notation

CTU : Cooperative Training Unit

GP1 : Graduation Project 1

GP2 : Graduation Project 2

IS : Information Systems

SDLC: Software Development Life Cycle

UI : User Interface

2FA : Two-Factor Authentication

IT : Information Technology

CS : Computer Science

GPA : Grade Point Average

WBS: Work Breakdown Structure

WSBLC : Waterfall System Development Life Cycle

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Chapter 1: Planning

1.1 Project Overview

Cooperative training plays an important role in bridging the gap between academic learning and real-world experience. However, students may face several challenges that prevent them from fully benefiting from this experience. One of the main challenges identified in this study is finding suitable training opportunities. The survey results showed that 51% of students faced difficulty finding organizations related to their majors. In a separate question, 43.1% of students reported that their training opportunities only partially matched their skills or interests, while 25.5% reported that their opportunities did not match their skills or interests (Appendix A).

The challenges are not limited to students. Interviews with Company A and Company B indicated that training organizations also face difficulties in finding suitable applicants who meet their requirements (Appendix D). In addition, the interview with the University Training Unit identified challenges related to placement management, student progress monitoring, and communication among stakeholders (Appendix C). Together, these findings indicate limitations in the current cooperative training process that may reduce its overall effectiveness.

These findings reveal a broader need for a well-organized and intelligent system that connects students, universities, and training organizations. This need is also supported by previous research on internship management systems [1]. The current cooperative training process relies on manual procedures, simple filtering, and basic platforms that may not adequately support student-opportunity matching, communication, and progress monitoring. Findings from the interviews further highlighted limitations in managing placements and communication between the involved stakeholders (Appendices C and D).

Artificial Intelligence (AI) provides an opportunity to improve this process by analyzing relevant data and identifying patterns that can support more informed placement decisions. In cooperative training, AI can analyze students' profiles, skills, academic performance, and interests together with training organizations' requirements to recommend more suitable opportunities.

To address the identified challenges, this project proposes the Smart Cooperative Training System (Awan), a centralized digital platform designed to improve the management of cooperative training. Awan will use AI-based matching to recommend suitable training opportunities based on students' qualifications and training organizations' requirements. The system will also support communication and progress monitoring among the involved stakeholders, providing a more organized and efficient cooperative training process.

1.2 Problem Statement

In an ideal cooperative training environment, students should be able to find training opportunities that align with their academic background, skills, and interests. Universities should be able to monitor students' progress throughout the training period, while training organizations should be able to identify suitable candidates based on their requirements. In addition, clear and continuous communication should be maintained among students, universities, and training organizations. An effective cooperative training process should therefore provide appropriate placement, organized communication, continuous monitoring, and clear coordination among all involved parties.

However, the current cooperative training process does not fully achieve this ideal situation. The findings of this study reveal several challenges related to finding suitable training opportunities, matching students with appropriate placements, communication, and progress monitoring. The survey results show that students experience difficulties in securing suitable training opportunities. In particular, 49% of the surveyed students required one to three months to secure a cooperative training opportunity, while 29.4% required more than three months. In addition, 54.9% reported applying to more than seven training organizations before receiving an acceptance (Appendix A). The interviews also identified challenges in communication and progress monitoring, as well as difficulties faced by training organizations in identifying suitable candidates (Appendices C and D).

These challenges demonstrate a gap between the ideal cooperative training environment and the current process. Students may struggle to find suitable opportunities, universities may face difficulties in monitoring students and maintaining effective communication, and training organizations may experience difficulties in identifying candidates who meet their

requirements. Therefore, there is a need for a more integrated and intelligent approach that can improve student-placement matching, communication, coordination, and progress monitoring throughout the cooperative training process.

1.3 Project Impact

The Awan system will be a centralized digital platform connecting students, universities, and training companies, representing significant improvements in the management, coordination, and general performance of cooperative training programs. The platform offers an organized and effective approach to handle all aspects of the training process, from posting opportunities and student applications to communication and progress monitoring, removing the need for manual processes and various systems.

The main impact of the system is to enhance the student placement process with the use of artificial intelligence. The system can analyze student profiles (skills, academic background, and career interests) and match them with training organizations' requirements and generate accurate recommendations of suitable opportunities. This smart matching method minimizes mismatches and raises the chances of students being placed in employment that match their course of study. In turn, this gives students more valuable practical experience that helps their academic development and increases their readiness for the job market.

The approach also directly deals with issues indicated through survey data, as many students reported difficulty in identifying relevant training options, often applying to many organizations, and long periods before receiving acceptance (According to Appendix A) . The approach decreases the work and time it takes for students to find a place by providing structured recommendations and a better overview of what is available.

AWAN also enhances communication and coordination through the integration of a centralized communication channel and a real-time notification system. Therefore, students, supervisors and training organizations are always informed about changes,

deadlines and choices. Moreover, the system offers an improved method of monitoring the students' performance as the students can provide weekly reports in a structured way which can be viewed by the university and company supervisors. This promotes transparency, decreases the need for manual verification and enables more accurate and efficient evaluation of student development.

1.4 Project Stakeholders

Stakeholders are individuals or groups who are interested in the project and are directly or indirectly affected by the system and its outcomes. Identifying the stakeholders is important because it clarifies their roles and responsibilities in supporting the successful implementation and operation of the system.

1.4.1 System Administrators (Intended Customer):

University Training Unit:

The University Training Unit is responsible for managing the system. It manages the cooperative training program, manages student and organization records, reviews and approves training opportunities, supervises overall training activity, and maintains the functionality of the system. The unit also facilitates communication between students, supervisors, and training organizations.

1.4.2 End Users of the System:

Cooperative Training Students:

Students represent the main users of the system. Through the platform, they are able to search for available training opportunities, submit applications, receive notifications, and monitor their training progress and tasks that they need to complete.

1.4.2.1 Training Organizations:

Training organizations also use the system to publish the available training opportunities, review and evaluate student applications, choose suitable trainees, and even provide feedback and evaluations on the performance of the students during the internship.

1.4.2.2 University Supervisors:

The platform allows academic supervisors to track the progress of students throughout their cooperative training, evaluate submitted reports, assess student performance, and contact students or training organizations when needed.

1.4.2.3 Company Supervisors:

Company supervisors are responsible for direct supervision of students in the training organization. They monitor the student's daily work, give feedback on performance, review submitted weekly reports, and communicate with university supervisors as needed to guarantee correct training progress and accurate evaluation.

1.5 Objectives

AWAN system will help to improve the organization of cooperative training programs by offering a smart platform connecting students, University training unit and training organizations. The system aims to enhance the training process by making sure that there is the right alignment between the skills, qualifications, and the training opportunities of the students.

The project objectives are classified into two, which are the objectives of AWAN system and the objectives of the academic objectives of the graduation project.

1.5.1 Awan System Objectives

• Reduce the time students spend looking for potential training opportunities by providing AI-based personalized recommendations.
• Improve student opportunity alignment using AI-based recommendations based on student skills and needs.
• Improve communication among all stakeholders by providing a centralized platform with messaging and notification features.
• Reduce time and effort for collecting and evaluating reports by providing a centralized report submission platform

1.5.2 Graduation Project Objectives

- Apply knowledge gained from information systems courses to identify cooperative training system problems and propose an academic solution.
- Design system architecture and main features based on information systems knowledge gained from IS courses.
- Enhance cooperation, communication, and research abilities during the project.

1.6 Approach

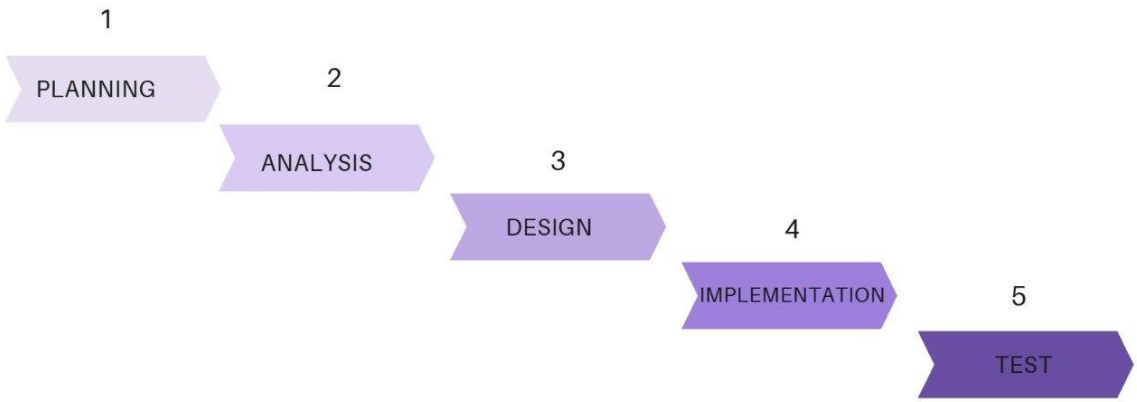


Figure 1Waterfall System Development Life Cycle[5]

Awan system adopts the Waterfall System Development Life Cycle (SDLC) as its development methodology because the project problem, objectives, and system requirements are clearly defined from the early stages. This makes the Waterfall model suitable for the project, as it allows the team to follow a structured and sequential development process. Each phase is completed before moving to the next phase, which helps maintain consistency throughout the project [4].

Unlike iterative approaches, this methodology supports the nature of this project, as the system requirements are clearly defined and documented before the development phase

begins, reducing the need for major changes during development and allowing the team to follow a structured, linear development process [4].

1.6.1 Planning Phase

During this phase, the team focused on clearly defining the problem context based on real challenges faced by students, as supported by the collected survey data .

In addition, the team identified the data requirements for the AI-based matching feature. Existing datasets related to students' skills and training requirements will be cleaned and prepared for use. A survey will also be used to collect information about training organizations' requirements to improve the accuracy of the matching process [4].

1.6.2 Analysis Phase

The analysis stage focused on making sense about the distinction between the existing practices and the preferred solution. The team analyzed the weaknesses of the current systems that lead to inaccurate matching or efficient coordination and converted them to system requirements. The role of the stakeholders such as students, training organizations, supervisors and the training unit was evaluated in terms of interaction and expectation by the system. This stage had a direct influence on the direction of functionality of AWAN, including improved matching, simplified application procedures, and overall tracking of progress[4].

1.6.3 Design Phase

In this stage, the team focused on defining the system requirements based on the problems identified in the previous phase. The required system features were then determined to address these problems. For example, the matching feature was defined to improve the alignment between students and suitable training opportunities, while the monitoring feature was defined to help supervisors track students' progress during training. Therefore, each system feature was linked to a specific identified problem [4].

1.6.4 Implementation Phase

The GP2 implementation phase will be the one where the designed solution will be

converted into a working one. The development will center on the creation of key aspects of the site including creating profile of students and companies, post training, applying in training and matching students to training placements. The Waterfall model is structured and thus the development will be guided by well-defined requirements, eliminating confusion and lessening significant changes [4].

1.6.5 Testing and Deployment Phase.

Lastly, we will test the system to ensure that it can handle the initial issue that was found in the project. The tests will not only concern whether the system functions properly, but also whether it can enhance matching accuracy, make communication simpler, as well as to help with better monitoring. When the system is proven correct, it can be implemented to facilitate cooperative training processes in a more [4].

1.7 Project Scope

The system is intended for students in technology-related fields such as information systems, information technology, computer science, and related majors, as well as university training units, supervisors, and training organizations. It facilitates cooperative training process by providing key functionalities such as creation of student profile, posting of opportunity, AI based matching and recommendations, management of applications, communication between stakeholders, notifications and monitoring of performance through weekly report submissions.

The website allows training companies to post available opportunities, students to search for and apply for suitable positions and supervisors to track student progress. The system is managed by the university training unit that approves opportunities and is responsible for the complete training process.

The project does not provide for full job recruitment aside from cooperative training. It does not enable the integration with external systems (e.g. government platforms, internal university systems, etc.). The system offers decision support by using AI-based matching but does not fully automate placement decisions. In addition, the system does not handle

the academic allocation or registration of students for cooperative training. The university is responsible for determining which students are eligible for cooperative training and providing their lists and relevant academic information to the system.

1.7.1 Assumptions and Constraints:

1.7.1.1 Assumptions:

- It is assumed that all stakeholders are authenticated before proceeding with any actions.
- It is assumed all supervisors will have a created account provided by their admins.

1.7.1.2 Constraints:

- Opportunities can only be posted by training organizations and approved by the university training unit.
- The system assists in decision-making but does not automatically choose placements.

1.7.2 Boundaries and Limitations:

- The project relies on self-collected or sample datasets due to the lack of ready datasets specific to cooperative training.
- The system is limited to only technology opportunities.
- The system is limited to the Riyadh region.

1.7.3 Deliverables:

- Proposal summary form.
- First meeting form.
- First progress report form .
- Second progress report form .
- Mid-term review form .
- Final report.
- Final presentation.

1.8 Work Breakdown Structure

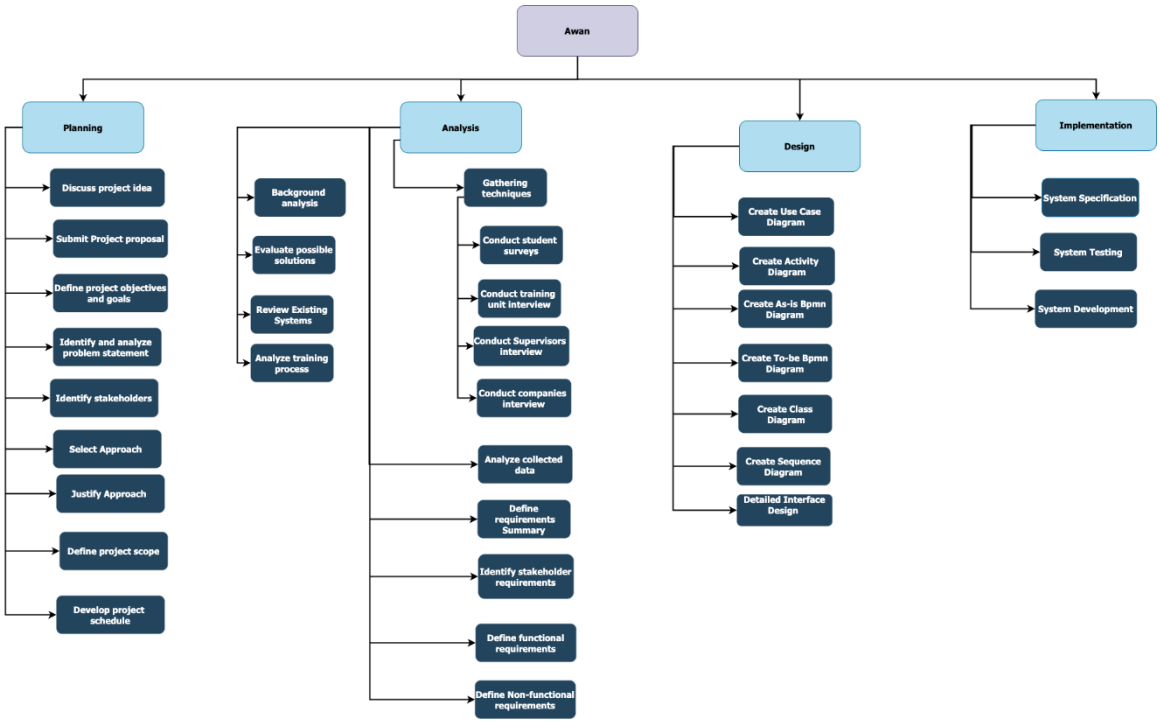


Figure 2 WBS

1.9 Gantt Chart (Time Frame)

Planning:

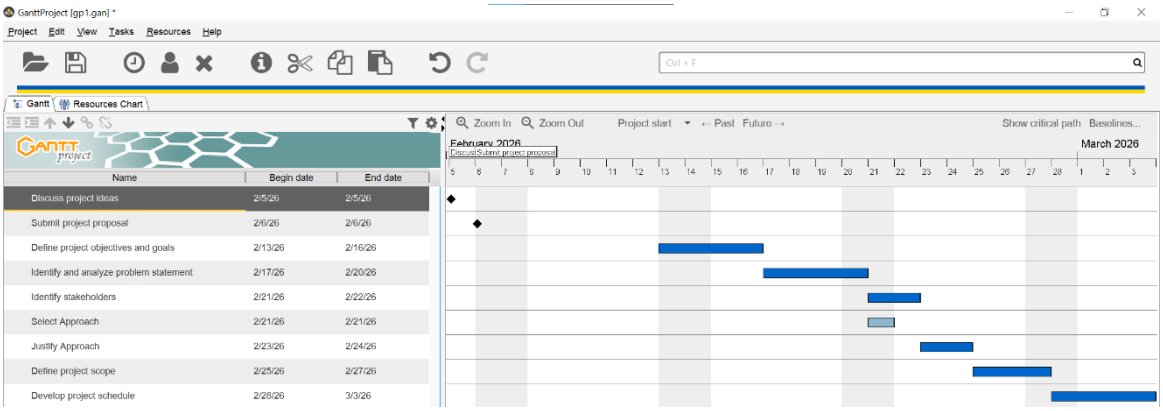


Figure 3GanttChart Planning

Analysis:

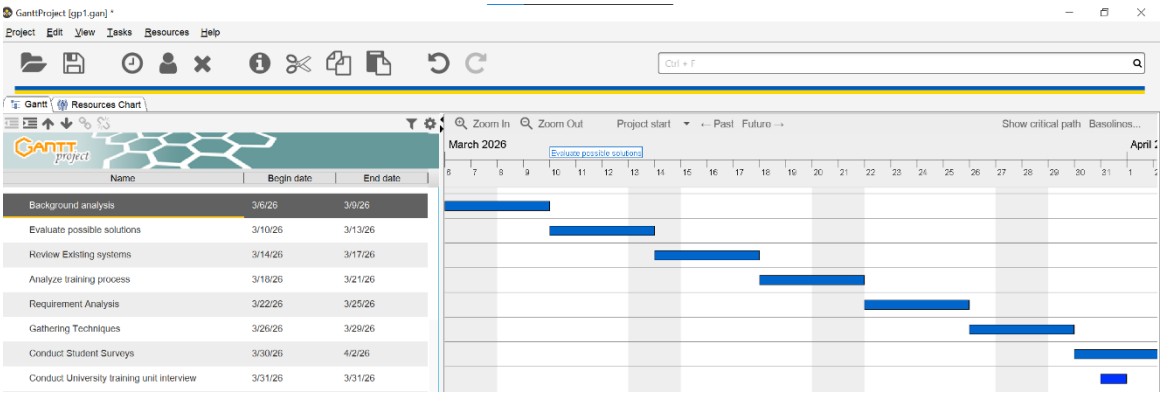


Figure 4GanttChart Analysis1

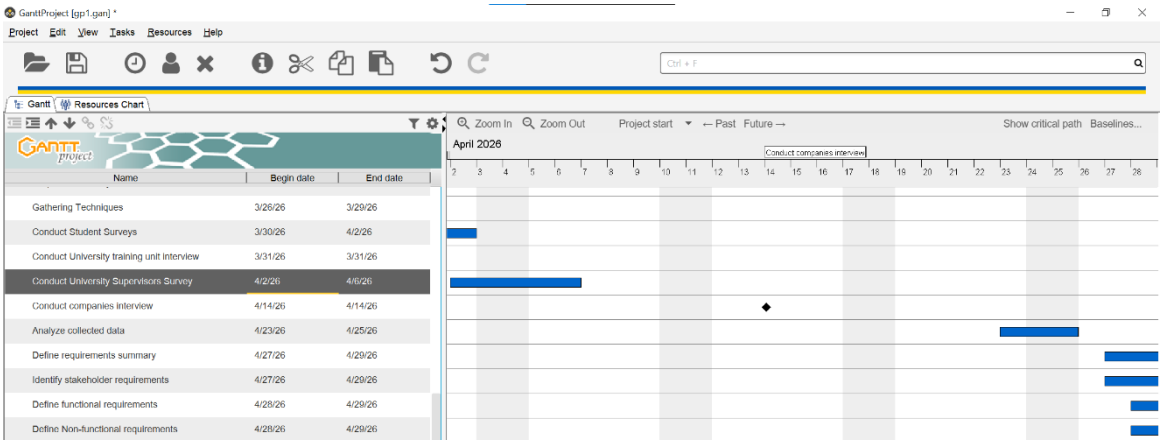


Figure 5GanttChart Analysis2

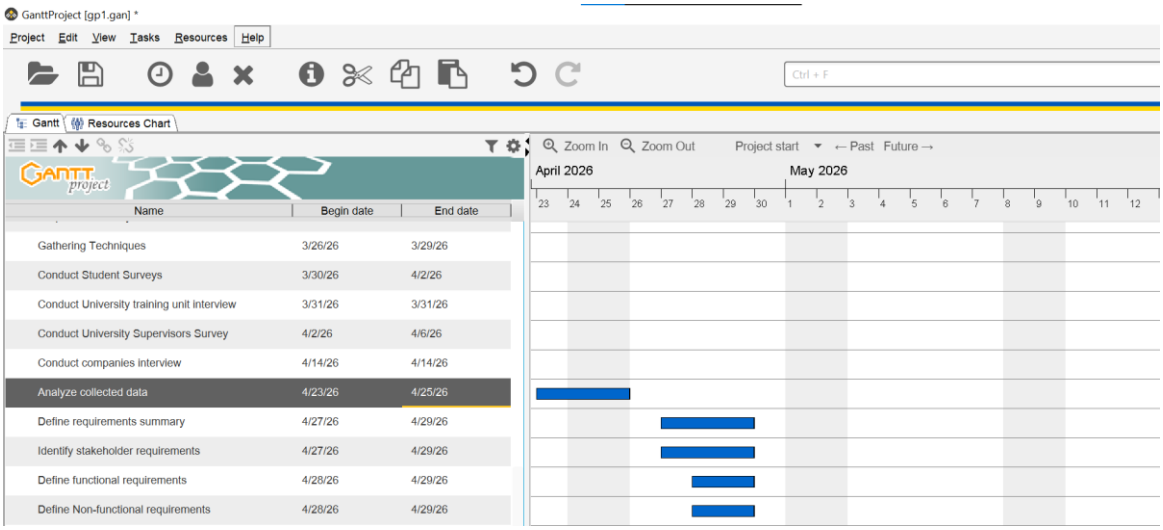


Figure 6GanttChart Analysis 3

Design:

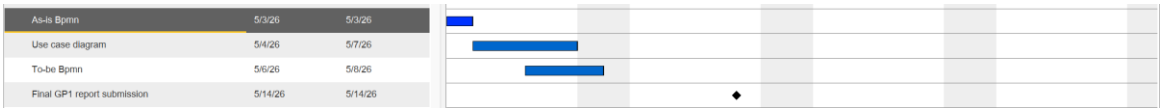


Figure 7GanttChart Design1

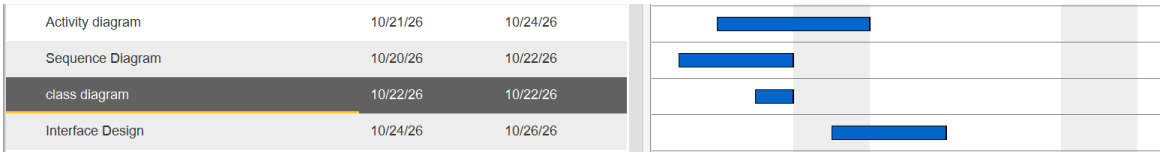


Figure 8 GanttChart Design2

Implementation:

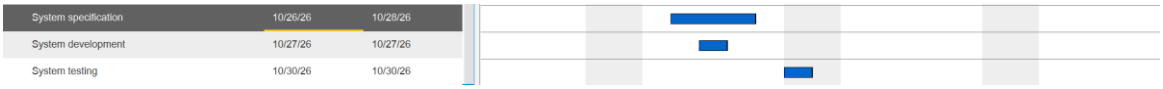


Figure 9GanttChart implementation

Chapter 2: Background Analysis

2.1 Possible Solutions

The team members have two practical solutions to develop the proposed Awan system, either a basic system without AI or an intelligent system with AI integration.

2.1.1 Basic system (without AI):

This solution covers the implementation of the project as a central website where students can find and browse various training programs and apply directly through the system. It provides the basic functionalities such as listing of opportunities, application processes, notifications, and communication between students, supervisors, and training organizations. This way, the process is made more accessible and organized for all stakeholders through different devices such as desktops, tablets, and mobile phones.

Users can login and use the system according to their roles. Students manually search for chances and apply for the same. However, the matching procedure relies mostly on simple criteria like GPA, major or keywords, which may not always provide accurate or appropriate results.

Therefore, the major problem of mismatch between students and training organizations may persist, therefore decreasing the overall effectiveness of the training experience and restricting the quality of suggestions supplied to users.

From the development point of view, this solution is easier to realize because it does not involve advanced technology or complex processing. It requires lower cost of development, easier maintenance and less training of users with adequate system performance and usability. However, this system is easier to build but still is affected by limited scalability and long-term effectiveness due to its inability to intelligently adapt to the needs of users or enhance the accuracy of matching over time.

2.1.2 Intelligent System (AI-enabled - AWAN) :

This solution involves the implementation of the system with AI integration that provides smart and automatic matching of students to training possibilities. The algorithm collects data from students about their abilities, GPA, courses they have finished and interests and compares that to the needs of the training organizations to come up with a matching percentage and then offer the most suited options. Instead of having to manually search across various opportunities, individuals get personalized recommendations that make decision making better, minimize search time and offer a more smooth and productive user experience.

The system also provides real-time performance tracking, automated notifications of missing reports or minimal participation and increased communication for all stakeholders through a centralized messaging and notification system. Such features increase the transparency, coordination and monitoring efficiency and allow supervisors to observe student progress more effectively.

This solution has a larger development cost, more technical complexity and stronger security considerations due to the integration of AI, real-time tracking, and the management of sensitive data from multiple stakeholders. These features require additional development effort and stronger data protection measures. However, it has better long-term value and scalability. The intelligent system can enhance the accuracy of matching with more data and be more successful in solving the core problem of mismatch between students and training companies, enhancing the overall quality and efficiency of cooperative training programs.

2.1.3 Cost-Benefit Analysis

Table 1

Criteria	Basic System (Without AI)	Intelligent System (With AI)
Development Cost	Less cost	More cost
Implementation Time	Short	Medium
Ease of Development	Yes	Moderate
Efficiency	Limited	High
User Experience	Moderate	High
Scalability	Limited	High
Long-term Value	Low	High

2.1.4 Final Decision

According to the analysis, the team chose the Intelligent System (With AI – Awan) solution as it delivers a more accurate and effective solution to the problem and gives a greater long-term value The solution addresses the identified problems by improving student-opportunity matching and supporting communication and progress monitoring within one centralized system..

Based on the cost-benefit analysis in Table3 , The team chose the Web Application method for the AI-based Awan system based on two perspectives:

Users' point of view: The web-based application offers access from any device, without any limitations, which makes it more versatile and appropriate for all stakeholders. It works on both desktop and mobile devices, so you may use it in any environment.

Developers' point of view: The web application saves development time and expense as

there is no need to design distinct versions for each OS. It also makes maintenance easier and removes delays tied to app store approvals, making it the most practical and effective alternative.

2.2 Overview of Existing systems

Before developing a new system like AWAN, it is necessary to study the existing similar systems, to collect the requirements of a project. It assists in highlighting advantages and disadvantages of existing solutions.

2.2.1 Ataba System

Ataba is a platform that provides training opportunities for students. However, its functionality is limited in terms of interaction and system features [6]. Ataba advantages are:

- Provides a list of available training opportunities for students.
- Simple and easy-to-use interface.

Ataba disadvantages are:

- Limited filtering options for searching opportunities.
- Redirects users to external websites for application, which affects user experience.
- No tracking system to monitor application progress.
- No integration with universities or supervisors.
- No communication features between stakeholders.
- Does not support AI-based recommendations or performance monitoring.
- No Student performance monitoring.

2.2.2 Taawoni System

Taawoni is a platform designed to facilitate cooperative training by connecting students with training opportunities and organizations [7].

Taawoni advantages are:

- Provides a structured database of training opportunities.
- Includes a built-in application process.

- Supports partnership with universities.
- Allows companies to participate in training and recruitment activities.
- Taawoni disadvantages are:
- Lacks advanced filtering capabilities for opportunities.
- No real-time notifications or status updates for applications.
- Limited tracking system without real-time alerts.
- No direct communication between stakeholders.
- No AI-based recommendation system.
- No Student performance monitoring.

2.2.3 University Field Training System

The University Field Training System provides basic services to support students during the field training process. The system mainly focuses on providing training-related forms and documents required during different stages of the training process.

The University Field Training System advantages are:

- Provides training forms for students to download.
- Allows students to issue a field training letter.
- Provides a training commencement form.
- Provides a final evaluation form.

The University Field Training System disadvantages are:

- Limited to basic field training services and forms.
- Does not provide training opportunity discovery.
- Does not provide AI-based opportunity recommendations.
- Does not provide direct communication between stakeholders.
- Does not provide application tracking.

- Does not provide continuous student performance monitoring during the training period.

Table 2.2 summarizes a comparison of existing systems used for cooperative training management, Atabah [6], Taawoni [7], and University Field Training System[8] highlighting the strengths and limitations of each system in terms of opportunity discovery, application process, tracking system, university integration, company integration, AI-based recommendations, performance monitoring, and communication with stakeholders. This comparison provides valuable insights into the gaps that AWAN aims to address and the features it seeks to enhance and implement .

Table 2summarizes a comparison of existing systems [6][7]

Category	**Ataba**	**Taawoni**	**University Field Training System**
Opportunity Discovery	Basic list of opportunities with minimal filtering capabilities	Structured database with categorized opportunities but lacks advanced filtering capabilities.	None
Application Process	Redirects users to external company websites for application.	Built-in application workflow in the system with no status updates and notifications.	None
Tracking System	None	None	None

University Partnership And Supervisor access	No university cooperation , supervisor access to interact with students	Yes, Cooperated with Universities No supervisor access to interact with students	University-based system with supervisor involvement through third-party tools.
Company Interaction	No direct interaction with companies.	Yes, Companies are involved in recruitment and training activities but have no direct interaction.	No direct interaction with companies.
AI Opportunity Recommendation	None	None	None
Performance Monitoring	None	None	Available ,limited
Communication With Stakeholders	No direct communication	No direct communication.	No direct communication.

After analyzing the mentioned systems, Ataba [6] and Taawoni [7] mainly support training opportunity discovery and application-related activities, while the University Field Training System [8] focuses on basic field training services. However, the analyzed systems do not provide a fully integrated solution that combines application tracking, continuous performance monitoring, effective communication between stakeholders, and AI-based opportunity recommendations.

AWAN team is committed to sustaining these benefits and addressing the drawbacks noted. The proposed solution aims to create a more seamless and convenient platform for improving the overall training experience for students, universities, and companies.

In the proposed system, AWAN will provide advanced opportunity discovery through smart filtering and AI-enabled recommendations to help students find suitable training opportunities efficiently. Further, AWAN puts special emphasis on strong integration with universities and companies and allows supervisors to actively monitor their students' progress and performance and communicate with them directly. The system also provides effective communication channels among all the stakeholders to enhance collaboration.

2.3 Existing Business Processes

As seen in Figure (10), the existing process in the Taawoni system begins when the student logs into the system. Once the student logs in, they select an action that they wish to take.

If the student wishes to change their information, they continue to update their information and exit. Or the student can see the status of the application, and then they can end the process.

If the student decides to look for opportunities, they will start by choosing their preferences and then the opportunities will be displayed. From there, they choose one of the opportunities and check the details of that opportunity. The student then makes the choice as to whether they would like to apply to the opportunity or not. If not, they return to 'browsing opportunities' again. If they decide to apply, the application is made then, and they are sent a confirmation e-mail message, the process ends.

On the company side, it starts by the company logging into the Taawoni system and choosing the desired action. The company can also review potential applications or post a new job opportunity.

Once the company decides to review the applications, it will show one student and they can review their application, and the process will end. Or they could simply look at applications without taking any steps.

When the company decides to post the opportunity, they fill the details and post it, then they upload a nomination test and training plan. After the above steps are taken, the process terminates. In general, the current process in Taawoni system has low integration among the steps and does not include advanced automation, such as communication activity and follow-up[7].

Taawoni As-is BPMN:

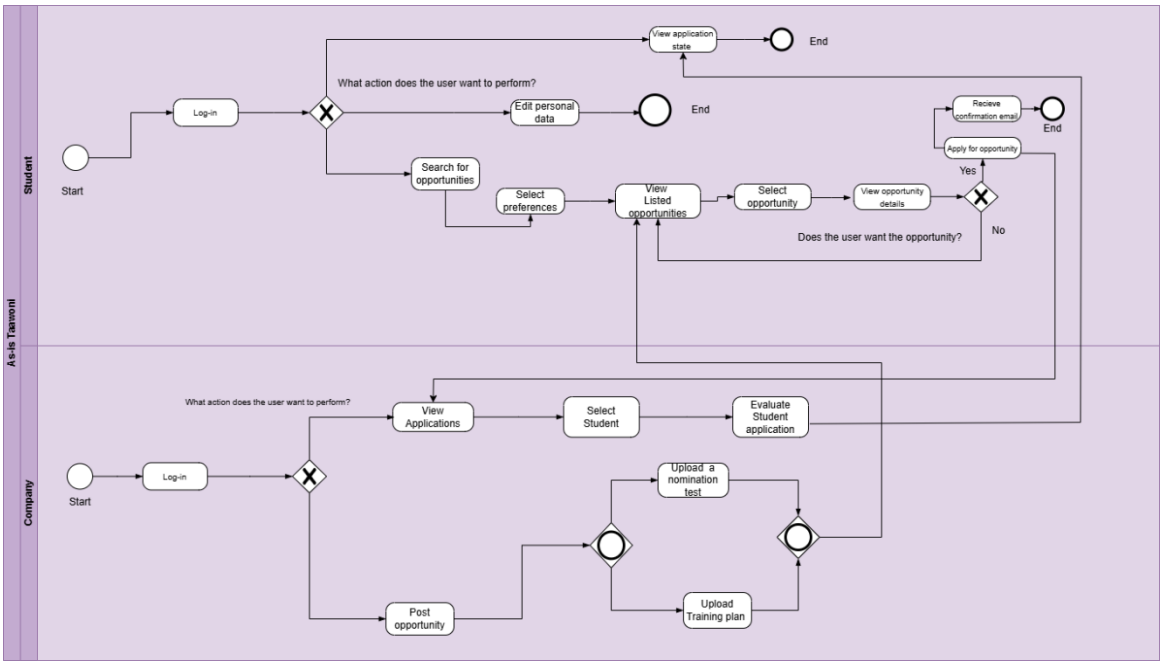


Figure 10 ai-is BPMN

As seen in Figure (10), the existing process in the Taawoni system begins when the student logs into the system. Once the student logs in, they select an action that they wish to take.

If the student wishes to change their information, they can edit their personal data and end the process. Alternatively, the student can view the application state and then end the process.

If the student decides to search for opportunities, they start by selecting their preferences, after which the available opportunities are displayed. The student can then select an opportunity and view its details. After reviewing the opportunity, the student decides whether to apply. If the student does not want to apply, they return to the listed opportunities. If they decide to apply, the application is submitted, and the student receives a confirmation email. The submitted application then becomes available to the company for review and evaluation.

On the company side, the process begins when the company logs into the Taawoni system and selects the desired action. The company can either view applications or post a new opportunity.

If the company decides to view applications, the system displays the received applications. The company then selects a student and evaluates the student's application. After the company evaluates the application, the application state is updated and can be viewed by the student through the “View Application State” option.

If the company decides to post a new opportunity, the opportunity is posted, and the company can upload a nomination test and a training plan. The posted opportunity then becomes available for students to view through the listed opportunities.

In general, the current process in the Taawoni system provides interaction between students and companies throughout the opportunity and application process. Companies can post opportunities that students can view and apply for, while companies can review and evaluate the submitted applications, and students can view their application state. However,

the process still has limited integration and does not include advanced automation for activities such as direct communication and continuous follow-up [7].

2.4 Literature Review

2.4.1 Introduction

Cooperative training plays an important role in connecting academic learning with practical work experience. However, managing the training process involves different stakeholders, including students, educational institutions, and training organizations. Internship Management Systems (IMS) can support this process by organizing internship-related activities and improving communication and coordination between the involved stakeholders [1][2].

Previous research has examined different approaches to improving internship management. Shatat et al. [1] studied the planning and organizational aspects of an Internship Management System, while Mydyti and Kadriu [2] examined how such systems can improve the relationship and communication between students, educational institutions, and businesses. These studies demonstrate the importance of using centralized systems to support and organize the internship process[1][2].

In addition to managing the training process, selecting suitable training opportunities is another important aspect of internship systems. Recent research has explored the use of intelligent technologies to improve internship matching. Mydyti and Ware [3] investigated the integration of intelligent web scraping, machine learning, and natural language processing to analyze student profiles and internship opportunities and improve the matching process based on factors such as competencies, interests, and professional objectives[3].

Therefore, this literature review examines previous research related to internship management systems, stakeholder communication and coordination, and intelligent internship matching. The reviewed literature provides a foundation for understanding the

approaches that can support the development of Awan and helps identify relevant concepts and technologies for improving the cooperative training process.

2.4.2 Internship Management Systems

2.4.2 Internship Management Systems

Internship Management Systems (IMS) are designed to organize and support the different activities involved in the internship process. These systems can provide a centralized environment where students, educational institutions, and training organizations can manage internship-related activities more efficiently. Shatat et al. [1] studied the planning phase of an internship management system and emphasized the importance of identifying system requirements and organizing the processes and interactions between the involved stakeholders before system development.

Mydyti and Kadriu [2] examined the role of internship management systems in improving the relationship between internship seekers, employers, and educational institutions. Their study showed that such systems can support communication and information exchange among the involved stakeholders and simplify internship-related processes. The study also highlighted the value of using a digital system to connect students with businesses and educational institutions throughout the internship process.

Overall, previous research indicates that internship management systems can provide a more organized and centralized approach to managing internship activities [1][2]. These findings are relevant to Awan, as the proposed system aims to bring students, training organizations, supervisors, and the university training unit together within one platform to support the cooperative training process.

2.4.3 Stakeholder Communication and Coordination

Good communication and coordination between the stakeholders are also essential aspects of the internship process. The student is required to communicate with educational institutions and training bodies, and the educational body or employer needs to share

information and arrange training activities. Handling internships and working together effectively with the parties involved can be more challenging if there is no organized way of communicating.

Mydyti and Kadriu [2] examined the relationship between internship seekers, employers, and educational institutions and highlighted the role of internship management systems in improving communication and interaction among these stakeholders. Their study showed that a digital internship management system can provide a common environment that facilitates communication and the exchange of internship-related information.

Similarly, Shatat et al. [1] considered the interaction between different stakeholders during the planning of an internship management system. Their work supports the importance of organizing the roles and processes of the parties involved within a centralized system.

These findings are relevant to Awan because the system involves students, training organizations, company supervisors, university supervisors, and the University Training Unit. Providing these stakeholders with a centralized platform can support communication and coordination throughout the cooperative training process.

2.4.4 Intelligent Internship Matching

Finding a suitable internship opportunity depends on how well the student's qualifications and interests match the requirements of the available opportunities. Traditional internship systems may provide access to opportunities, but intelligent technologies can support a more personalized matching process by analyzing information from both students and internship opportunities.

Mydyti and Ware [3] explored the integration of intelligent web scraping techniques into internship management systems to improve internship matching. Their study incorporated Machine Learning (ML) and Natural Language Processing (NLP) techniques to analyze internship opportunities and student profiles. The matching process considered factors such as students' competencies, interests, and professional objectives to identify more relevant internship opportunities.

The study demonstrates how intelligent technologies can enhance internship management systems by moving beyond basic opportunity searching toward more personalized recommendations [3]. This concept is particularly relevant to Awan, where Artificial Intelligence (AI) is used to recommend cooperative training opportunities based on student information and the requirements of training organizations. Therefore, intelligent matching can support students in identifying opportunities that are more closely aligned with their profiles and can also help training organizations identify suitable candidates.

2.4.5 Previous Studies and Relation to Awan

The reviewed studies address different aspects of internship management systems. Shatat et al. [1] focused on the planning and organization of an internship management system, emphasizing the importance of identifying system requirements and organizing the processes between stakeholders. Mydyti and Kadriu [2] focused more on improving the relationship and communication between internship seekers, employers, and educational institutions through an internship management system.

In comparison, Mydyti and Ware [3] focused on the intelligent aspect of internship management by exploring techniques that can improve the matching between students and internship opportunities. Their study demonstrates how intelligent technologies can be used to provide more relevant opportunities based on student profiles and internship information.

Awan builds on these different areas by combining internship management, stakeholder communication and coordination, and intelligent opportunity matching within one cooperative training system. The reviewed studies therefore provide a useful foundation for the main concepts and functions considered in the development of Awan.

2.4.6 Conclusion of the Literature Review

The literature review examined several important aspects related to cooperative training systems, including internship management, stakeholder communication and coordination, and intelligent internship matching. Shatat et al. [1] provided insights into the planning and organization of internship management systems, while Mydyti and Kadriu [2] highlighted

the importance of improving communication and relationships among students, employers, and educational institutions. Mydyti and Ware [3] demonstrated the potential of intelligent technologies to improve the matching between students and suitable internship opportunities.

Overall, the reviewed studies provide a foundation for the development of Awan. The findings support the need for an organized system that connects the involved stakeholders while also using intelligent matching to recommend suitable training opportunities. Awan combines these concepts to support a more organized and effective cooperative training process.

Chapter 3: Requirements Analysis

3.1 Requirement Gathering Summary Results

Requirement gathering is essential throughout system development process since project team must define and record functional and non-functional needs of the system. For this project, the team employed many approaches to obtain a full understanding of the cooperative training process, the requirements of stakeholders, and the major challenges students face along their training journey.

In this case, background analysis was conducted to examine current platforms for cooperative training and internship management such as University Field Training System ,Ataba and Taawoni to determine their strengths, limitations, and missing functions [6][7]. In addition, the relevant research on the internship management system, communication between stakeholders and the intelligent matching approach were studied to understand more about the existing practice and technology development of the cooperative training systems [1][2][3][8].

Moreover, a questionnaire was sent to students who went through cooperative training to collect quantitative data on the difficulties they encountered in the search for training options, communication with supervisors and completion of training requirements. The questionnaire was also useful to identify the expectations and views of students about the proposed system (According to Appendix A).

The project team interviewed employees from the Cooperative Training Unit (CTU)(According to appendix C), university supervisors, and representatives from training organizations to gather more detailed qualitative information about the current training process, communication methods, supervision processes, and the main problems during the training period (According to Appendix C and D). Interviews provided a better comprehension of stakeholder expectations and system requirements.

3.1.1 Background Analysis

The project team used background analysis as one of the key requirement collection methodologies to better understand the cooperative training area. This method contributed to identifying limitations and gaps in the current systems, as well as in the understanding of existing platforms for the management of cooperative training and internships. The analysis was based on a review of similar platforms, the services and features offered by them, together with their strengths and weaknesses.

The project team analyzed Ataba platform to better understand how training opportunities are displayed and managed for students. The review revealed that the platform is primarily focused on the advertising of chances through simple opportunity listings with limited filtering options and that students are referred to the companies' external websites during the application process. Moreover, the platform lacks tracking methods, supervisor access, direct stakeholder communication, AI-powered suggestions, and performance monitoring features [6].

The team also looked at Taawoni platform to see how they manage opportunities, the application process and the interaction between students and organizations. The analysis found that Taawoni offers a more structured database of opportunities and a combined application process. But the system does not have advanced filtering capabilities, tracking systems, direct contact between stakeholders, AI based recommendations, performance monitoring, supervisor involvement with students during the training session[7].

The team also analyzed the University Field Training System to understand the services provided by the university to support students during their field training. The analysis showed that the system mainly focuses on basic administrative training services, including downloading training forms, issuing field training letters, submitting training commencement forms, and submitting final evaluation forms. However, other activities, such as communication and student progress monitoring, depend on third-party tools and are not integrated into the system. In addition, the system does not provide opportunity discovery, an application process, application tracking, direct company interaction, or AI-based opportunity recommendations.

The study by Shatat et al. [1] was analyzed independently to better understand the planning phase and organizational structure of internship management systems. The study highlighted the need for centralized management, organized communication channels and enhanced cooperation between stakeholders throughout the internship process [1].

The study of Mydyti and Kadriu [2] was evaluated to understand the role of internship management systems in strengthening communication and connections between internship participants, employers, and universities. The study emphasized the need to improve the cooperation, availability of internship information and supervision processes during training [2].

Furthermore, the work of Mydyti and Ware [3] was studied gto get an idea of how intelligent technologies and smart matching methods might be integrated into internship management systems. The study showed how intelligent recommendation techniques can enhance the opportunity matching based on student qualifications, abilities and interests, which assisted the project team to gain a better understanding of the possible function of intelligent matching features in cooperative training systems [3].

Overall, the background study improved the knowledge about the current practices of cooperative training and aided in identifying numerous potentials of improvement that could contribute to the design and development of the proposed system.

3.1.1.1 Background Analysis Structure

The research team studied and evaluated three cooperative training platforms that support management of training opportunities and communication between the stakeholders. The analysis reviewed prior research on cooperative training management systems and internships, with a particular focus on University Field Training System, Ataba and Taawoni [8][6][7].

The comparison was based on key areas such as finding opportunities, application process, communication methods, supervisor access, company involvement, AI recommendation

features, and performance monitoring. Team members also actively compared these platforms and studies to assess the advantages and limitations of each system. Such comparative analysis offered a great insight into the existing challenges of students, supervisors and training companies in the cooperative training process.

Based on the background analysis, the main requirements identified for Awan system are:

- The company shall be able to communicate.
- The company supervisor shall be able to communicate .
- The university supervisor shall be able to communicate.
- The university training unit shall be able to communicate.
- The university training unit shall be able to view assigned students.
- The university supervisor shall be able to view assigned students.
- The company supervisor shall be able to view assigned students.

3.1.2 Questionnaire

A questionnaire was conducted on Wednesday, 4th of March 2026, to collect information on students' experiences during cooperative training and to determine the most frequent problems students encountered in their training journey. The questionnaire was administered to students who had previously had co-op training in technology-related majors. It was passed on through Google Forms using student communication channels such as WhatsApp groups and social media platforms(According to Appendix A).

The questionnaire comprised 11 closed-ended questions and had 52 responses (According to appendix A). The questions were on the accessibility of the training opportunities, communication with the supervisors, challenges in submitting reports, the suitability of the opportunities to the students' skills and interests and the overall success of the cooperative training process (According to Appendix A) .

The results of the questionnaire showed some significant conclusions on the existing cooperative training situation (According to Appendix A). Several students had trouble identifying appropriate training organizations where 43.1% described the process as

difficult and 37.3% as very difficult. Moreover, 49% of students reported to have spent one to two months to get a training opportunity and 29.4% needed more than three months (According to Appendix A).

Additionally, 54.9% of students applied to more than seven organizations before getting accepted, according to the statistics, highlighting the strong competition and limited availability of appropriate opportunities. Students also reported several challenges during the search phase, including slow responses from organizations (74.5%), competition with many students (56.9%), lack of training opportunities (54.9%), no centralized platform for training opportunities (52.9%) and difficulty to find organizations related to their majors (51%)(According to Appendix A).

In terms of training placement compatibility, 43.1% of students indicated their training was only partially suited to their skills and interests and 25.5% indicated opportunities did not at all correspond to their skills and interests. The study also revealed that 36% of students had difficulty in speaking with university supervisors during the training time and 32% partially had difficulty in communication (According to Appendix A). Furthermore, 28% of students said they had issues sending performance reports from training organizations to university supervisors and 36% reported partially facing difficulties in this process (According to Appendix A).

Based on the Questionnaire , the main requirements identified for Awan system are:

- The student shall be able to search for opportunities.
- The student shall be able to view recommended opportunities.
- The student shall be able to apply for opportunities.
- The student shall be able to manage reports.
- The student shall be able to receive notifications.
- The student shall be able to communicate.

3.1.2 Interview

The information was gathered through interviews to understand the cooperative training process, the activities of supervisors, the methods of communication and the expectations of the stakeholders. The interviews comprised participants from the Cooperative Training Unit (CTU), university supervisors and training organizations. The interviews were conducted through online meetings and direct communication sessions with open-ended and semi-structured questions(According to appendix B, C and D).

3.1.2.1 Cooperative Training Unit Interview

Two employees from the Cooperative Training Unit (CTU) were interviewed on Tuesday, 14th of April 2026 . The interviews included 16 open-ended and semi-structured questions about the cooperative training workflow, registration procedures, university policies, student follow-up strategies, and administrative problems during the training period(According to appendix B).

The results achieved during the interview gave a clear picture of the responsibilities of the CTU and the current workflow followed from the start of the training period until the end. The interviews also highlighted some of the existing limitations such as problems in managing communication among stakeholders, delays in responding by the organization, and lack of centralized administration of opportunities and follow-up of students(According to appendix B).

3.1.2.2 University Supervisors' Interviews

Two interviews were held with university supervisors, Dr. Moudi Alsheikh on Wednesday, 22nd of April 2026 and Dr. Mona Alkhattabi on Friday, 17th April 2026. The interviews included 22 open and semi-structured questions related to student supervision, communication techniques, performance monitoring, report evaluation and the main challenges supervisors face during the cooperative training period (As shown in appendix C) .

Interviews showed that supervisors had difficulties with the efficiency of communication, the follow-up procedures and the monitoring of students' development in a consistent way. The supervisors highlighted the need for enhancing the collaboration between university and organization supervisors through centralized communication and regular online follow-up approaches. They also emphasized the importance of more practical and planned training activities that are more consistent with student interests and preferences (According to Appendix C).

3.1.2.3 Company Interviews

Interviews were conducted with representatives from two training companies, referred to as Company A and Company B on Monday,11th of May 2026. Including one employee per company identified as Employee A and Employee B. For privacy and confidentiality reasons, the names of the companies and employees are not disclosed.

The interviews consisted of two kinds of questions. The interview with company supervisors contained 14 open-ended semi-structured questions, concentrating on their duties as a supervisor, evaluation of students, their interactions with the universities and tracking the progress of the training (According to appendix D). Trainee management interviews included 11 questions about opportunity management, student acceptance procedures, posting training opportunities, and organizational challenges during cooperative training (According to Appendix D).

The interview results pointed to the necessity for more direct communication between company managers and university supervisors, especially on the issues of attendance tracking, student evaluation and weekly follow-up. Participants also identified issues related to inaccurate student reporting and difficulty with monitoring completed assignments and proper attendance(According to Appendix D) . The companies also emphasized the need for formal evaluation procedures, such as standard evaluation forms or surveys that may be directly examined by university supervisors (According to Appendix D).

Finally, companies highlighted the importance of managing available seats for training opportunities, posting opportunities through a centralized platform and clearly assigning students to company supervisors while managing a specific limit of students for each supervisor to reduce the disorganization and improve training efficiency. They also underlined the importance of having a more structured procedure that organizations can use to examine student applications and clearly accept or reject students through a centralized approach, as opposed to doing it manually(According to Appendix D).

Based on the Interviews, the main requirements identified for Awan system are:

- The company shall be able to upload opportunities.
- The company shall be able to manage applicants.
- The company shall be able to assign a supervisor.
- The company shall be to receive notifications.
- The company supervisor shall be able to evaluate students.
- The company supervisor shall be able to receive notifications.
- The university supervisor shall be able to evaluate students.
- The university supervisor shall be able to receive notifications.
- The university training unit shall be able to approve opportunities.
- The university training unit shall be able to assign supervisors.
- The university training unit shall be able to receive notifications.

3.2 Stakeholder Requirements

AWAN system involves several key stakeholders, including students, companies, company supervisors, university supervisors, and the university training unit. Each stakeholder interacts with the system in different ways based on their role. The following section outlines the requirements for each stakeholder.

3.2.1 Student Requirements

3.2.1.1 The student shall be able to search for opportunities.

3.2.1.2 The student shall be able to view recommended opportunities.

3.2.1.3 The student shall be able to apply for opportunities.

3.2.1.4 The student shall be able to manage reports.

3.2.1.5 The student shall be able to receive notifications.

3.2.1.6 The student shall be able to communicate.

3.2.2 Company Requirements

3.2.2.1 The company shall be able to upload opportunities.

3.2.2.2 The company shall be able to manage applicants.

3.2.2.3 The company shall be able to assign a supervisor.

3.2.2.4 The company shall be to receive notifications.

2.2.2.5 The company shall be able to communicate.

3.2.3 Company Supervisor Requirements

3.2.3.1 The company supervisor shall be able to view assigned students.

3.2.3.2 The company supervisor shall be able to evaluate students.

3.2.3.3 The company supervisor shall be able to communicate .

3.2.3.4 The company supervisor shall be able to receive notifications.

3.2.4 University Supervisor Requirements

3.2.4.1 The university supervisor shall be able to view assigned students.

3.2.4.2 The university supervisor shall be able to evaluate students.

3.2.4.3 The university supervisor shall be able to communicate.

3.2.4.4 The university supervisor shall be able to receive notifications.

3.2.5 University Training Unit Requirements

3.2.5.1 The university training unit shall be able to approve opportunities.

3.2.5.2 The university training unit shall be able to assign supervisors.

3.2.5.3 The university training unit shall be able to view assigned students.

3.2.5.4 The university training unit shall be able to receive notifications.

3.2.5.5 The university training unit shall be able to communicate.

3.3 Proposed Business Process

The proposed business process (to-be) shown in Figure 11 illustrates how the Awan system supports and manages the cooperative training process through one centralized platform that connects students, universities, supervisors, companies, and the AI recommendation system. The process begins when companies upload training opportunities through the platform. The uploaded opportunities are then reviewed by the University Training Unit, which decides whether to accept or reject the opportunity based on the provided information. If the opportunity is approved, it becomes available for students to search and apply for. If rejected, the company receives the rejection result and can review the opportunity information again if needed.

Students can search for available opportunities manually or request AI-based recommendations. The AI system analyzes both student information and opportunity information simultaneously to generate matching recommendations and display suitable results to students. After reviewing opportunities or recommendations, students can apply

for suitable opportunities through the platform and later track their application status and registration information.

On the company side, organizations can review submitted applications and decide whether to accept or reject applicants. If an applicant is accepted, the company may assign a company supervisor to monitor the student during the training period. The system then updates the related student assignment information. In parallel, the University Training Unit can view registered students and assign university supervisors responsible for following up with students academically during the training period.

Both university and company supervisors can access assigned student information through the platform. Supervisors can monitor student progress and review updates. Also, they can evaluate student performance throughout the cooperative training period. It additionally allows students to submit their reports and check their evaluations and updates from the supervisors.

Moreover, the platform also improves communication and coordination by providing users with application status, assignment status, approvals, evaluation, and registration information. The process also supports continuous monitoring and follow-up for all stakeholders involved throughout the cooperative training journey.

3.3.1 Awan To-be BPMN:

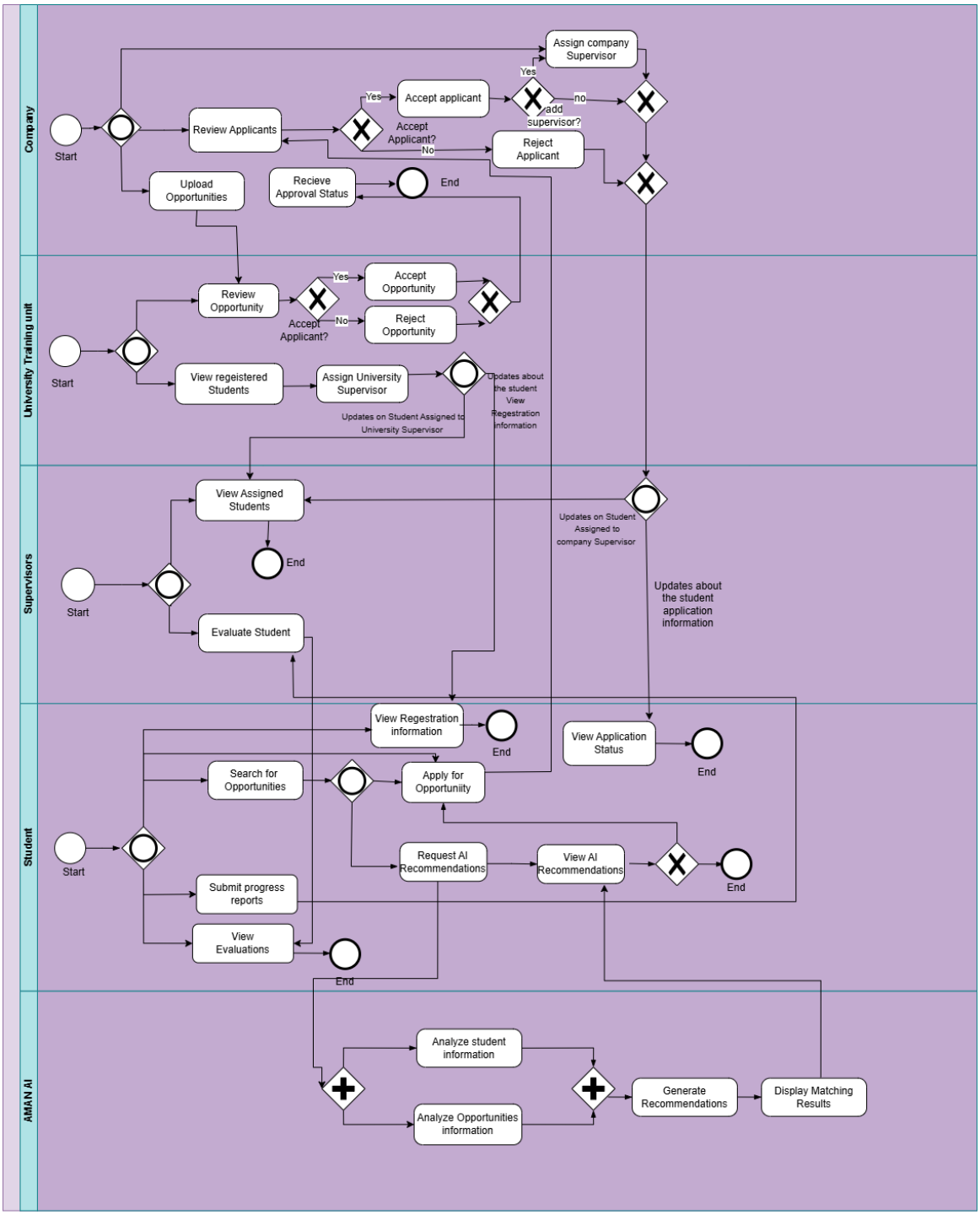


Figure 11To-Be BPMN

Overall, as shown above the proposed business process creates a more organized and connected cooperative training environment by improving opportunity management, student follow-up, communication, evaluation procedures, and recommendation accuracy

while supporting the entire training process from opportunity posting until student evaluation and report submission.

3.4 Functional Requirements

In this section, the project team members identified the functional requirements that the Awan website must perform. Team members utilized the use case diagram and the use case descriptions.

3.4.1 Use-Case Diagram

The use case diagram reflects the main functions for Awan website. Figure 12 shows the actors associated with the use cases they are included in, and the relationships of the use cases to each other.

Awan Use Case Diagram:

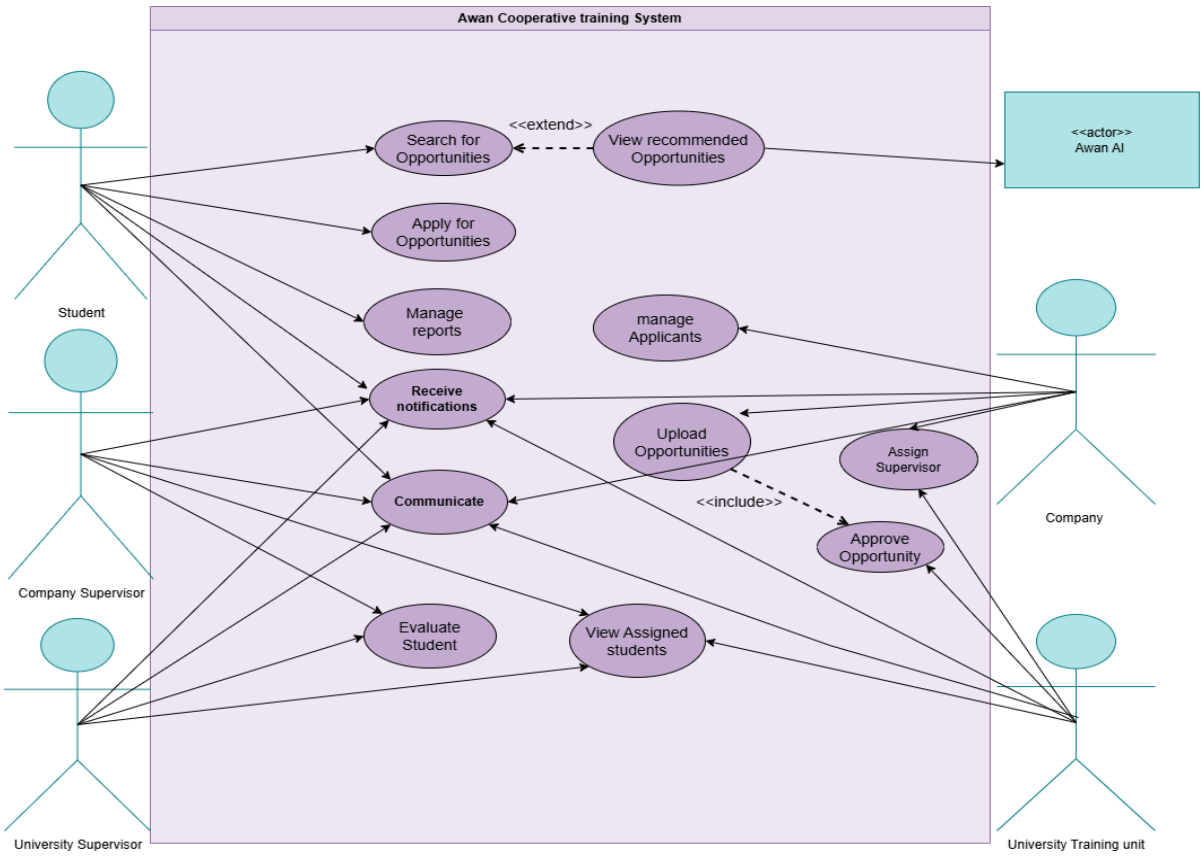


Figure 12Use Case Diagram

3.4.2 Use Case Descriptions

The written description of how stakeholders perform each use case and tasks in the website. As shown in Tables 3 to 14

Table 3 Search for Opportunities Description

<table>
<tr><td>Use Case Name: Search for Opportunities</td><td colspan="2">ID:1</td><td>Importance Level: High</td></tr>
<tr><td colspan="2">Primary Actor: Student</td><td colspan="2">Use Case Type: Detail, Essential</td></tr>
<tr><td colspan="4">Brief Description: This use case describes how a student searches for available cooperative training opportunities through the system.</td></tr>
<tr><td colspan="4">Trigger: The student selects the “Search for Opportunities” option from the system.
Type: External</td></tr>
<tr><td colspan="4">Relationships:
Association: Student, Awan AI
Extend: View Recommended Opportunities</td></tr>
<tr><td colspan="4">Normal Flow of Events:
1. The student logs into the system.
2. The student selects the “Opportunities”.
3. The system displays the search feature.
4. The student chooses either normal search or AI search.
-If the student selects AI search, go to “View Recommended Opportunities” use case.
-If the student selects normal search, continue to step 5.
5. The student enters keywords or search filters.
- If the student leaves the search field empty go to Alternative Flow 5a.
- If valid keywords or filters are entered, continue to step 6.</td></tr>
</table>

6. The system searches for opportunities based on the entered keywords or selected filters. - If no matching opportunities are found, go to Exception Flow 6a. - If matching opportunities are found, continue to step 7. 7. The system displays matching opportunities. The student views the search results. - If the student wants to choose the opportunity go to" Apply for Opportunities" use case - If the student does not want to choose the opportunity end the flow
Alternate Flows: **5a. Empty Search Input** 1. The student leaves the search field empty without selecting filters. 2. The system requests the student to enter keywords or select filters. 3. Continue to step 5 in the Normal Flow. **Exceptional Flows:** **6a. No Matching Opportunities Found** 1. The system displays a message indicating that no matching opportunities were found. 2. The student modifies the keywords or search filters and continues to step 5 in the Normal Flow or leave.

Table 4Apply for Opportunities

Use Case Name: Apply for Opportunities	**ID:2**	**Importance Level: High**
Primary Actor: Student	**Use Case Type: Detail, Essential**	
Brief Description: This use case will describe how the student applies for a training opportunity through the system.		

Trigger: The student selects an opportunity and clicks on “Apply”.

Type: External

Relationships:

Association: Student

Normal Flow of Events:

.

1. The students view available opportunities from the Opportunities section.
2. The student selects an opportunity to either view its details or apply for it.
    - If the student chooses to view the opportunity details, continue to step 3.
    - If the student chooses to apply directly, continue to step 4.
3. The system displays the opportunity details
    - If the student decides to choose it, go to step 4.
    - If the student does not want it, go to step 1.
4. The student clicks “Apply”.
    - If the student decides not to continue with the application process, go to Alternative Flow 4a.
    - If the student confirms the application, continue to step 5.
5. The system checks whether the student has already applied for the same opportunity.
    - If the student has already applied, go to Exception Flow 5a.
    - If the student has not applied before, continue to step 6.
6. The system checks whether the student profile information is complete.
    - If the profile information is incomplete, go to Alternative Flow 6a.
    - If the profile information is complete, continue to step 7.
7. The system submits the application.
8. The system confirms the successful submission.
9. The student receives a confirmation message.

Sub-Flows:
Alternative Flows: **4a. Application Cancelled** 1. The student cancels the application process before submission. 2. The system returns the student to the opportunity page. 3. The use case ends. **6a. Incomplete Profile Information** 1. The system detects missing required profile information. 2. The system requests the student to complete the missing information. 3. The student updates the required profile information. 4. Continue to step 8 in the Normal Flow. **Exceptional Flows:** **5a. Student Already Applied** 1. The system displays a message indicating that the student has already applied for this opportunity. • If the student wants to complete search goes to flow 1 Search for Opportunities step 3 • If the student wants to cancel end the flow.

Table 5Upload Opportunities

Use Case Name: Upload Opportunities	**ID:3**	**Importance Level: High**
Primary Actor: Company	**Use Case Type: Detail, Essential**	
Brief Description: This use case describes how the Company uploads training opportunities to the system.		

Trigger: The Company selects the "Upload Opportunities" feature.

Type: External

Relationships:

Association: Company.

Extend: Approve Opportunity

Normal Flow of Events:

1. The company logs into the system.
2. The company selects "Opportunities" section.
3. The system displays the system with the "Post New Opportunity" option.
4. The company selects "Post New Opportunity".
5. The company fills in the required information regarding the opportunity.
6. The company clicks "Submit for Approval".
    - If the required information is incomplete, go to Alternative Flow 6a.
    - If the submission process fails due to a system error, go to Exception Flow 6b.
    - If the information is complete and the submission is successful, continue to step 7.
7. The system redirects the opportunity to the "Approve Opportunity" use case.
8. The company receives the approval response.
    - If the opportunity is approved, continue to step 9.
    - If the opportunity is rejected, go to Alternative Flow 8a.
9. The system publishes the opportunity successfully.
10. The system displays a confirmation message to the company.

Alternate / Exceptional Flows:

Alternative Flows:

6a. Incomplete Opportunity Information

1. The system requests the company to complete the missing required information.
2. The company updates the missing information.
3. Continue to step 6 in the Normal Flow.

8a. Opportunity Rejected

1. The system notifies the company that the opportunity was rejected.
2. The company chooses either to edit and resubmit the opportunity or cancel the process.
    - If the company chooses to edit and resubmit, continue to step 5 in the Normal Flow.
    - If the company chooses to cancel, the flow ends.

Exceptional Flows:

6b. Submission Failure

1. The system fails to submit the opportunity due to a system error.
2. The system displays an error message to the company.
3. The company retries the submission process by going back to step 4

Table 6Assign Supervisor

Use Case Name: Assign Supervisor	**ID:4**	**Importance Level: High**
Primary Actor: University Training unit, Company		**Use Case Type: Detail, Essential**
Brief Description: This use case describes how the University Training unit and the Company assign supervisors to a student during the training period.		

Trigger: The University Training unit or Company selects the "Assign Supervisor" option **Type**: External
Relationships: Association: University Training unit, University Supervisor
Normal Flow of Events: 1. The University Training unit or Company logs into the system. 2. The system displays the "Assign Supervisor" option. 3. The University Training unit or Company selects "Assign Supervisor". 4. The system displays the list of supervisors. 5. The University Training unit or Company checks whether a supervisor already exists in the system. • If the supervisor does not exist, go to Alternative Flow 5a. • If the supervisor already exists, continue to step 6. 6. The University Training unit or Company selects a supervisor. 7. The system displays the list of students. 8. The University Training unit or Company selects a student. • If the University Training Unit or Company wants to view the student's information before assignment, go to Sub-Flow 8a. • Otherwise, continue to step 9. 9. The system checks whether the selected supervisor can supervise additional students. • If the supervisor has reached the maximum number of assigned students, go to Alternative Flow 9a. • If the supervisor can supervise additional students, continue to step 10 10. The University Training unit or Company assigns the selected supervisor to the student. • If the assignment process fails due to a system error, go to Exception Flow 10a. • If the assignment is successful, continue to step 11. 11. The system stores the supervisor's assignment information.

12. The system updates the student assignment information successfully.
13. The system displays a confirmation message.

Sub-Flows:

8a. View Student Information

1. The University Training unit or Company selects a student profile.
2. The system displays the student information and training details.
3. Continue to step 9 in the Normal Flow.

Alternate / Exceptional Flows:

Alternative Flows:

5a. Supervisor Does Not Exist

1. The University Training unit or Company adds a new supervisor to the system.
2. The system stores the supervisor's information successfully.
3. Continue to step 6 in the Normal Flow.

9a. Supervisor Reached Maximum Limit

1. The system displays a message indicating that the supervisor has reached the maximum number of assigned students.
2. The University Training unit or Company selects another supervisor.
3. Continue to step 9 in the Normal Flow.

Exceptional Flows:

10a. Assignment Process Failed

1. The system fails to assign the supervisor due to a system error.
2. The system displays an error message and requests the user to try again later.
3. Continue to step 10 in the Normal Flow.

Table 7Manage Applicants

Use Case Name: Manage Applicants	**ID:5**	**Importance Level: High**
Primary Actor: Company		**Use Case Type: Detail, Essential**
Brief Description: This use case describes how the Company reviews and accepts student applications for training opportunities.		
Trigger: The Company accesses submitted student applications. **Type**: External		
Relationships: Association: Company		
Normal Flow of Events: 1. The Company logs into the system. 2. The Company selects Applicants section 3. The Company selects an applicant from the recent applicants list. 4. The system displays the student application details with available actions, including "Accept Applicants". 5. The Company reviews the student application details. 6. The Company chooses to accept, reject, or request an interview for the applicant. • If the Company rejects the applicant, go to Alternative Flow 6a. • If the Company requests an interview, go to Alternative Flow 6b. • If the Company accepts the applicant, continue to step 7. 7. The system checks whether there are available seats for the opportunity. • If no seats are available, go to Alternative Flow 7a. • If seats are available, continue to step 8. 8. The system updates the application status as accepted.		

9. The system updates the number of available seats for the training opportunity. 10. The system sends an acceptance notification to the student.
Alternate / Exceptional Flows: **Alternative Flows:** **6a. Applicant Rejected** 1. The system updates the application status as rejected. 2. The system sends a rejection notification to the student. 3. The use case ends. **6b. Interview Requested** 1. The Company enters the interview details. 2. The system updates the application status as interview requested. 3. The system sends an interview notification to the student. 4. Continue to step 6 in the Normal Flow. **7a. No Available Seats** 1. The system displays a message indicating that no seats are available for the opportunity. 2. The Company selects another decision for the applicant. 3. Continue to step 6 in the Normal Flow.

Table 8 Receive notifications

Use Case Name: Receive notifications	**ID:6**	**Importance Level: Medium**
Primary Actor: Student, Company, Company Supervisor, University Supervisor, University Training unit		**Use Case Type: Detail, Essential**

Brief Description: This use case describes how users receive notifications related to applications, opportunities, supervisor assignments, messages, and training updates.

Trigger: The system generates a notification after an action or update occurs.

Type: External

Relationships:

Association: Student, Company, Company Supervisor, University Supervisor, University Training unit

Normal Flow of Events:

1. An action or update occurs in the system.
2. The system generates a notification.
3. The system determines the related recipient.
4. The system sends the notification to the related user.
    - If the notification delivery fails, go to Exception Flow 4a.
    - If the notification is delivered successfully, continue to step 5.
5. The user receives the notification on the dashboard or in the notifications section.
6. The user selects the notification.
7. The system displays the notification details.
8. The system will sign the notification as a readied.

Sub-Flows:

Alternate / Exceptional Flows:

Exceptional Flows:

4a. Notification Delivery Failed

1. The system fails to send the notification.
2. The system retries the notification delivery or displays an error message.

3. Continue to step 4 in the Normal Flow.

Table 9 Send Messages

Use Case Name: communicate	**ID:7**	**Importance Level: High**
Primary Actor: Student, Company Supervisor, University Supervisor, University Training unit	**Use Case Type: Detail, Essential**	
Brief Description: This use case describes how users send messages and communicate with each other through the system.		
Trigger: The user selects the "Send Messages" feature. **Type**: External		
Relationships: Association: Student, Company, Company Supervisor, University Supervisor, University Training unit		
Normal Flow of Events: 1. The user logs into the system. 2. The user selects the "Send Messages" section. 3. The system displays the list of available contacts or conversations. 4. The user selects a recipient or opens an existing conversation to reply to a received message. • If the user selects multiple recipients, go to Sub-Flow 4a. • If a single recipient or conversation is selected, continue to step 5. 5. The user enters the message content.		

- If the message content is empty, go to Alternative Flow 5a.
- If the message content is valid, continue to step 6.

6. The user sends the message.
7. The system delivers the message to the selected recipient.
    - If the message delivery fails, go to Exception Flow 7a.
    - If the message is delivered successfully, continue to step
8. The recipient receives a message notification.

Sub-Flows:

4a. Select Multiple Recipients

1. The user selects multiple recipients from the contacts list.
2. The system adds the selected recipients to the message conversation.
3. Continue to step 5 in the Normal Flow.

Alternate / Exceptional Flows:

Alternative Flows:

5a. Empty Message Content

1. The user attempts to send an empty message.
2. The system requests the user to enter message content before sending.
3. Continue to step 5 in the Normal Flow.

Exceptional Flows:

7a. Message Delivery Failed

1. The system fails to deliver the message.
2. The system displays an error message and requests the user to try again later.
3. Continue to step 6 in the Normal Flow.

Table 10 Manage Reports

Use Case Name: Manage Reports	**ID:8**	**Importance Level: High**
Primary Actor: Student	**Use Case Type: Detail, Essential**	
Brief Description: This use case describes how a student manages training reports by submitting new reports, viewing previous submissions, and viewing upcoming submissions.		
Trigger: The student selects the "Manage Reports" section. **Type**: External		
Relationships: Association: Student		
Normal Flow of Events: 1. The student logs into the system. 2. The student selects the "Manage Reports" section. 3. The system displays the available options: "Submit New Report", "View Previous Submissions", and "View Upcoming Submissions". 4. The student selects an option. • If the student selects "View Previous Submissions", the system displays previously submitted reports with their feedback status. • If the student selects "View Upcoming Submissions", the system displays upcoming submissions with their due dates. • If the student selects "Submit New Report", continue to step 5. 5. The system displays the available reports. 6. The student selects the required report and completes the corresponding report form. 7. The student may upload supporting files. • If the student uploads supporting files, go to Sub-Flow 7a. • Otherwise, continue to step 8. 8. The student submits the report. • If required information is incomplete, go to Alternative Flow 8a.		

• If the report submission fails, go to Exception Flow 8b. • If the report is submitted successfully, continue to step 9. 9. The system stores the report and updates its status as submitted. 10. The report becomes available to the university supervisor for review and evaluation. 11. The system displays a confirmation message. 12. The system redirects the report to the "Evaluate Student" use case. 13. The student may return later to view the report grade and feedback.
Sub-Flows: **7a. Upload Supporting Files** 1. The student selects a file to upload. 2. The system uploads and attaches the file to the report. 3. Continue to step 8 in the Normal Flow.
Alternate / Exceptional Flows: **Alternative Flows:** **8a. Incomplete Report Information** 1. The system requests the student to complete the missing required information. 2. The student completes the missing information. 3. Continue to step 8 in the Normal Flow. **Exceptional Flows:** **8b. Report Submission Failed** 1. The system fails to submit the report due to a system error. 2. The system displays an error message and asks the student to try again. 3. Continue to step 8 in the Normal Flow.

Table 11View Assigned Students

Use Case Name: View Assigned Students	**ID: 9**	**Importance Level: High**

Primary Actor: University Supervisor, Company Supervisor		**Use Case Type: Detail, Essential**
Brief Description: This use case describes how supervisors view students assigned to them through the system.		
Trigger: The supervisor selects the "View Assigned Students" feature. **Type**: External		
Relationships: Association: University Supervisor, Company Supervisor		
Normal Flow of Events: 1. The supervisor logs into the system. 2. The supervisor selects the "View Assigned Students" feature from the dashboard or menu. 3. The system retrieves the assigned student information. • If no students are assigned to the supervisor, go to Alternative Flow 3a. • If assigned students are found, continue to step 4. 4. The system displays the list of assigned students. 5. The supervisor selects a student. 6. The system displays the selected student details and progress information. 7. The supervisor chooses to either evaluate the student or send a message. • If the supervisor chooses to evaluate the student, the system redirects to the "Evaluate Student" use case. • If the supervisor chooses to send a message, the system redirects to the "Send Messages" use case. • If no additional action is selected, the use case ends.		

Alternate / Exceptional Flows: **3a. No Assigned Students** 1. The system does not find any assigned students for the supervisor. 2. The system displays a message indicating that no students are currently assigned. 3. The use case ends.

Table 12 Evaluate Student

Use Case Name: Evaluate Student	**ID:10**	**Importance Level: High**
Primary Actor: University Supervisor, Company Supervisor	**Use Case Type: Detail, Essential**	
Brief Description: This use case describes how the Supervisors evaluates the student progress		
Trigger: The supervisor selects the "Evaluate Student" feature. Type: External		
Relationships: Association: University Supervisor, Company Supervisor		
Normal Flow of Events: 1. The supervisor logs into the system. 2. The supervisor selects the "Evaluate Student" section, "Assigned Students", or "View All Reports" from the dashboard. • "Assigned Students" is used to access reports for a specific student. • "View All Reports" is used to access submitted reports for review and evaluation.		

3. The system displays the related students and submitted reports.
4. The supervisor selects a student or report for evaluation.
5. The system displays the student's submitted reports, uploaded files, and evaluation details.
    - If the required report has not been submitted, go to Alternative Flow 5a.
    - If the required report is available, continue to step 6.
6. The supervisor decides whether to download the uploaded files for review.
    - If the supervisor chooses to download the files, go to Sub-Flow 6a.
    - Otherwise, continue to step 7.
7. The supervisor reviews the student's performance and enters the evaluation details and feedback.
8. The supervisor submits the evaluation.
    - If the supervisor wants to modify the evaluation before submission, go to Sub-Flow 8a.
    - If the evaluation submission fails, go to Exception Flow 8b.
    - If the evaluation is submitted successfully, continue to step 9.
9. The system stores the evaluation information successfully.
10. The system notifies the student about the evaluation update.

Sub-Flows:

6a. Download Uploaded Files

1. The supervisor selects the uploaded files.
2. The system downloads the selected files.
3. Continue to step 7 in the Normal Flow.

8a. Edit Evaluation

1. The supervisor modifies the evaluation before submission.
2. The system updates the evaluation information.
3. Continue to step 8 in the Normal Flow.

Alternate / Exceptional Flows: **Alternative Flow** **5a. Required Report Not Submitted** 1. The system indicates that the required report has not been submitted. 2. The supervisor returns to the list of students or reports. 3. The use case ends. **Exception Flow** **8b. Evaluation Submission Failed** 1. The system fails to submit the evaluation due to a system error. 2. The system displays an error message and asks the supervisor to try again. 3. Continue to step 8 in the Normal Flow.

Table 13 View Recommended Opportunities

Use Case Name: View Recommended Opportunities	**ID:11**	**Importance Level: High**
Primary Actor: Student	**Use Case Type: Detail, Essential**	
Brief Description: This use case describes how the student views recommended training opportunities generated by Awan AI based on the student profile, skills, interests, and preferences.		
Trigger: The student selects the "View Recommended Opportunities" feature. **Type**: External		
Relationships: Association: Student, Awan AI		
Normal Flow of Events: 1. The student selects the "AI Recommendations "option .		

2. The system retrieves the student profile information, skills, interests, and preferences.
    - If the student information is incomplete, go to Alternative Flow 2a.
    - If the student information is complete, continue to step 3.
3. The system sends the student information to Awan AI.
    - If the system fails to connect with Awan AI, go to Exception Flow 3a.
    - If the connection is successful, continue to step 4.
4. Awan AI analyzes the student information and matches it with available opportunities.
5. Awan AI returns the recommended opportunities to the system.
    - If no matching opportunities are found, go to Alternative Flow 5a.
    - If recommended opportunities are found, continue to step 6.
6. The system displays the recommended opportunities to the student.
7. The student views the recommended opportunity details.
8. To continue and apply for opportunity go to "Search for Opportunities" , use case flow number 7

Alternate / Exceptional Flows:

Alternative Flows:

2a. Incomplete Student Information

1. The system requests the student to complete the missing required information.
2. The student updates the missing information.
3. Continue to step 1 in the Normal Flow.

5a. No Matching Opportunities Found

1. The system displays a message indicating that no recommended opportunities are available.

2. The student may return to the normal search option.
    - If the student wants to complete search go to “search for opportunities” ,case flow 5
    - If the student does not want to complete search end the flow

Exceptional Flows:

3a. AI Connection Failed

1. The system fails to connect with Awan AI.
2. The system displays an error message and requests the student to try again later.
3. Continue to step 1 in the Normal Flow.

Table 14 Approve Opportunity

Use Case Name: Approve Opportunity	**ID:12**	**Importance Level: High**
Primary Actor: University Training unit	**Use Case Type: Detail, Essential**	
Brief Description: This use case describes how the University Training unit reviews and approves uploaded training opportunities before they are published in the system.		
Trigger: The University Training unit selects the “Approve Opportunity” feature. Type: External		
Relationships: Association: University Training unit		

Normal Flow of Events:

1. The University Training unit logs into the system.
2. The University Training unit selects the "Review Opportunity Requests" section.
3. The system displays the pending opportunity requests submitted by companies.

• If there are no pending opportunity requests, go to Alternative Flow 3a.

• If pending opportunity requests are available, continue to step 4.

4. The University Training unit selects an opportunity request.
5. The system displays the opportunity details.
6. The University Training unit reviews the opportunity information.
7. The University Training unit decides whether to approve or reject the opportunity request.

• If the opportunity request is rejected, go to Alternative Flow 7a.

• If the opportunity request is approved, continue to step 8.

8. The University Training unit selects the "Approve Opportunity" option.
9. The system updates the opportunity status to approved.
10. The system publishes the opportunity successfully.
11. The system sends a notification to the Company regarding the approval decision.
12. The process returns to the "Review Opportunity Requests" section to review other pending requests.

Alternate / Exceptional Flows:

Alternative Flows:

3a. No Pending Opportunity Requests

1. The system displays a message indicating that there are no pending opportunity requests available.
2. The use case ends.

7a. Opportunity Rejected

1. The University Training unit rejects the opportunity request.

2. The system updates the opportunity status to rejected.
3. The system sends a rejection notification to the Company.
4. The process returns to the “Review Opportunity Requests” section.

3.5 Non-functional Requirements

Non-functional requirements specify the quality attributes and operational requirements of AWAN system. These requirements are designed to guarantee a stable and performance-efficient system for students, supervisors, and training systems, aiming for system usability, security, reliability, and system availability.

3.5.1 Performance Requirements

3.5.1.1 The system shall display search results and process training opportunity applications within 5 seconds under normal operating conditions.

3.5.1.2 The system should be able to handle at least 50 concurrent users without noticeable delays or performance issues.

3.5.2 Availability Requirements

3.5.2.1 The system must operate 24 hours per day, seven days per week.

3.5.2.2 System maintenance should be performed during non-peak hours (between 12 a.m. and 4 a.m.).

3.5.3 Security Requirements

3.5.3.1 The system shall ensure user authentication through secure credentials provided by the email and the password to log in.

3.5.3.2 The system shall use a strict password protocol that requires uppercase letters, lowercase letters, numbers and special characters.

3.5.3.3 Users should access information and functions based on their assigned role (Student, Supervisor and/or Organization).

3.5.3.4 The system should implement 2-factor authentication (2FA) using verification codes sent to users' phone numbers or email addresses.

3.5.4 Usability Requirements

3.5.4.1 The system's user interface shall be easy to navigate, allowing students, supervisors, and training providers to access the main features easily.

3.5.4.2 The system should give clear instructions and feedback messages to assist users in completing tasks.

3.5.5 Reliability Requirements

3.5.5.1 The system shall handle unexpected errors without losing user data and shall display an appropriate error message to the user.

3.5.6 Compatibility Requirements

3.5.6.1 The system must be able to be accessed with the most popular web browsers (e.g. Google Chrome, Microsoft Edge, Safari).

3.5.6.2 The system shall provide access via desktop computers, tablets and smartphones.

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Appendix

A. Miscellaneous

Appendix A

Student Survey questions:

Link:
https://docs.google.com/forms/d/e/1FAIpQLSfyoGuM1_ITtidWprutRtVN8axoJEy_bL9CH-jrW5B8X5U2oA/viewform?usp=header

Q1: How difficult was it for you to find a training organization?



Figure 13Q1: How difficult was it for you to find a training organization?

Q2: How long did it take you to find a training opportunity?



Figure 14 Q2: How long did it take you to find a training opportunity?

Q3:How many organizations did you apply to before getting accepted?

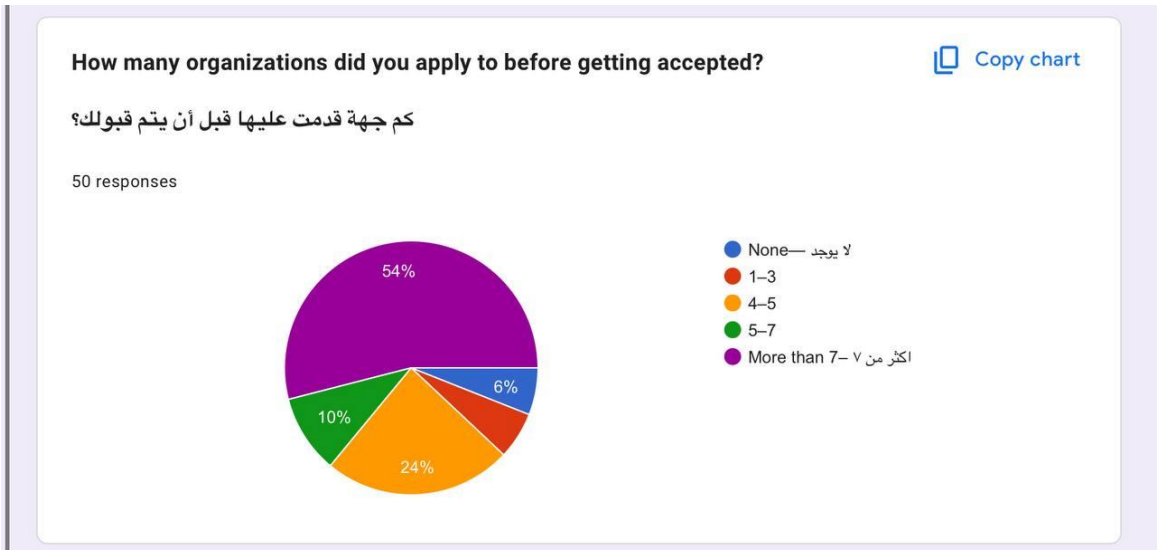


Figure 15How many organizations did you apply to before getting accepted?

Q4: What challenges did you face while searching for a training opportunity?



Figure 16 What challenges did you face while searching for a training opportunity?

Q5: Do you think the training you found matched your skills and interests?



Figure 17 Do you think the training you found matched your skills and interests?

Q6: Was finding a training opportunity easy with clear options or guidance available?



Figure 18 Was finding a training opportunity easy with clear options or guidance available?

Q7: Did you face difficulty communicating with your university supervisor during the cooperative training ?

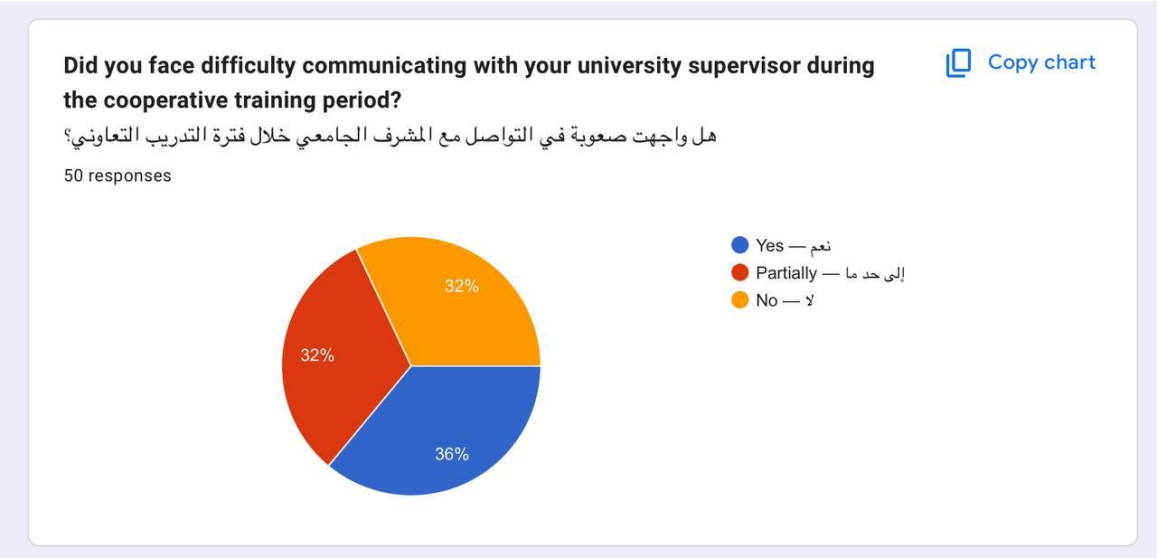


Figure 19 Did you face difficulty communicating with your university supervisor during the cooperative training ?

Q8: Did you face any difficulty sending your performance reports from the training organization to your university supervisor?

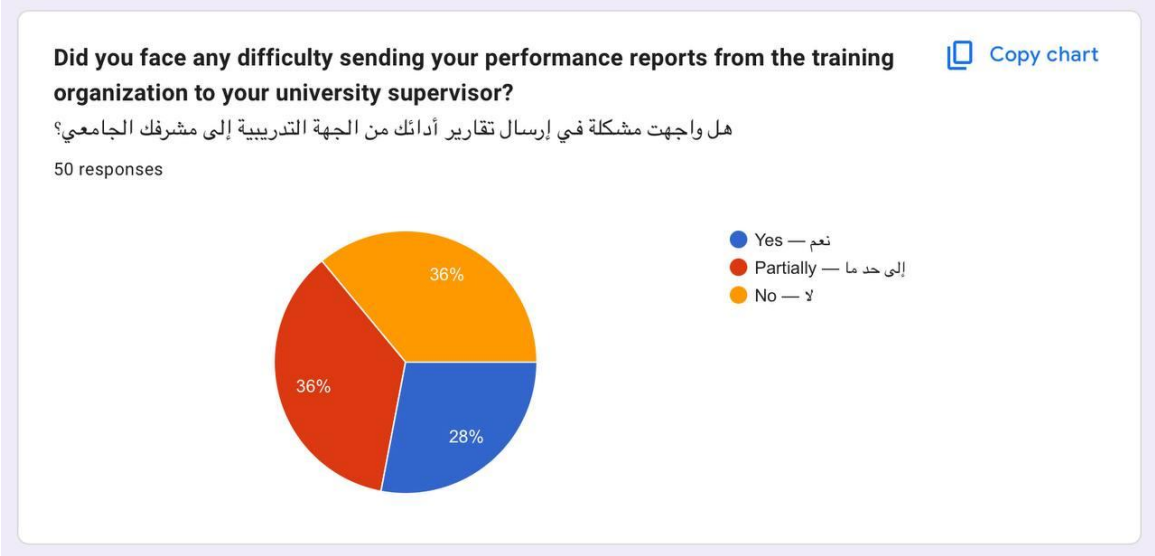


Figure 20 Did you face any difficulty sending your performance reports from the training organization to your university supervisor?

Q9: Do you think having a cooperative training platform would make the training process more efficient?



Figure 21 Do you think having a cooperative training platform would make the training process more efficient?

Q10: Do you have any suggestions to enhance the system?

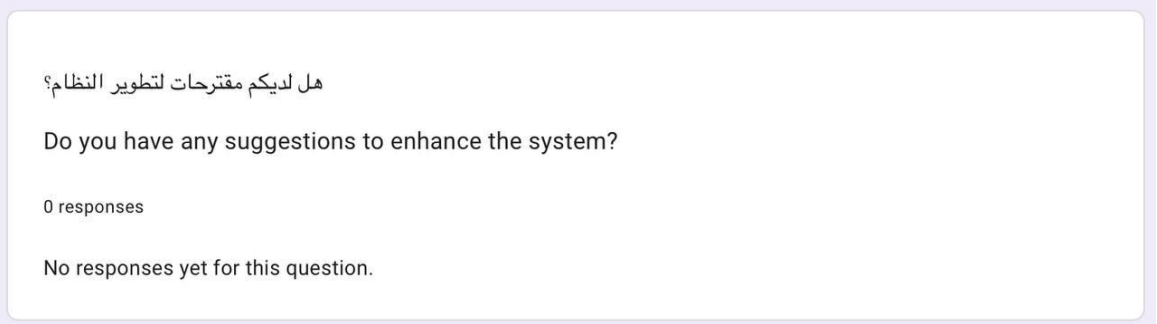


Figure 22 Do you have any suggestions to enhance the system?

Q11: Did you face any other problems during your cooperative training period?



Figure 23 Did you face any other problems during your cooperative training period?

Appendix B

Cooperative training unit interview questions:

First: Cooperative Training Committee Role and Responsibilities

1- What is your current role in supporting students during the cooperative training period?

They said they helped students from the beginning of the training till the end, from placing students into sections until they get a Cooperative Training Commencement Letter and get an approval from the company about the opportunity they applied to.

2- How do you communicate with students before, during, and after the training?

Through blackboard, email, sometimes phone numbers

3- What are the main services you provide to students at this stage?

Helping them get allocated to a certain opportunity, and adding to the approved opportunities list if isn't already there

4- How do you prepare students before they start their training?

Make sure they have completed all required courses and are at the stage to start coop training.

Second: Understanding Current Challenges

5- What are the most common problems students face during cooperative training?

They stated students may not find opportunities in time to start training, Sometimes they get approval to start a month or two later. Or they find difficulty in finding an opportunity in time.

6- Is there difficulty in matching students with suitable training organizations?

They said this depended on the student, but they try to find the opportunities that best suit their major.

7- What challenges do you face as a committee in monitoring students?

Sometimes student don't submit reports in time or don't provide feedback in time. As well as problems reaching this student when it happens.

8- Do students usually meet the expectations of training organizations?

Not always

Third: Communication with Training Organizations

9- How do you coordinate with companies or organizations that host trainees?

They said they had a list of companies that they send students to apply to or students could bring their own preference of a company.

10- Are there specific criteria for selecting training organizations?

Based on the training plan provided

11- Do students face difficulties in finding enough training opportunities?

They stated there are a lot of Opportunities but it's hard for students to find suitable training.

12- How do you handle issues or complaints from training organizations?

They don't have a way of communication with organizations.

Fourth: Evaluation and Follow-up

13- How do you monitor students' performance during training?

The supervisor tracks their performance, or they get an email in case of any issues escalated.

14- Do you use a specific system to evaluate the training process?

Blackboard and banner

15- How do you collect feedback from students or training organizations?

They take feedback from the students through a survey , if the organization isn't suitable they remove it from the list of approved opportunities.

16- How often do you follow up with students during their training period?

Only at the beginning and the end, or they may answer any inquiries they have.

Fifth: Improvement and Development

17- What aspects of the cooperative training system would you like to improve?

Direct Communication with organizations, a proper way to monitor students.

18- If a digital system (e.g., an application) were available, what key features would you expect? Direct communication with supervisors, either from the company or university, and with students, a centralized platform for managing report submissions and for students to apply to opportunities on this same platform.

19- How can we make it easier for you to manage and monitor students?

By the same way mentioned before, allowing students to submit reports and apply for opportunities in one platform.

Sixth: System/Application-Related Questions

20- What key data or reports do you need on a regular basis?

The Cooperative Training Commencement Letter and the final evaluations of students

21- Is it important for you to have automated notifications or reminders for students?

Yes, because students don't always submit reports and letters on time. Also, their attendance in such opportunities needs to be monitored.

Closing Question

22- Is there anything we haven't mentioned that you think is important for improving the cooperative training experience?

No

Appendix C

University Supervisors' questions

M.alkhattabi, personal communication, April 17 2026.

Dr.Mona Alkhattabi Meeting Confirmation

M.alsheikh, personal communication, April 22 2026.

Dr.Moudhi Alsheikh Meeting Confirmation

Supervisor 1

-How are students assigned to you as a supervisor?

They are assigned by the university department.

-Is the assignment based on students' preference?

No.

-Are you assigned to supervise more than one student?

Yes, I supervise multiple students. each supervisor is assigned to section, and each section has a number of students.

-What information do you receive about each student at the beginning of the training?

Basic information such as name , university ID number, and other basic information.

-How do you currently communicate with your students?

Through email or WhatsApp.

-How often do you communicate with students during the training period?

Weekly or when needed.

-Is there communication between you and the university supervisor?

Yes.

- how does this communication take place?

Mostly through email.

-How do students submit their weekly reports?

Through Blackboard.

-In what format are the reports submitted?

PDF or Word.

-What information should each report include?

Each report should clearly describe the tasks completed during the week, including what the student worked on and their specific responsibilities. It should also highlight key achievements, such as completed deliverables, skills gained, or any successful outcomes

-How do you review and evaluate the reports?

I review the submitted reports, download each student's file, and then evaluate their performance by assigning grades based on the quality and completeness of their work.

-How do you track student progress over time?

I track student progress through the submitted reports, especially by reviewing the reports collected toward the end of the training period to assess overall development.

-Do you keep records of student reports and performance? If yes, how?

Yes, I keep records as digital files by saving students' reports and evaluation results electronically.

-What challenges do you face when managing multiple students?

Managing a large number of students is challenging, especially when I have to download and review each student's file individually for evaluation.

-Is it difficult to track all student submissions and feedback?

Yes, sometimes it can be difficult to keep track of all submissions and feedback.

-How do you perform the final evaluation of students?

The final evaluation is based on the student's overall performance, submitted reports, and the feedback from company supervisors.

-What criteria do you use for evaluation?

The evaluation is based on students' commitment, the quality of their work, and their overall progress throughout the training period.

-What is the biggest difficulty you face during the supervision process?

The main difficulty is handling a large number of students and reviewing each report manually

-What are the main stages students go through during the training period?

The main stages include application, training, submitting reports, and final evaluation.

-How many field visits are usually conducted during the training?

Usually, one or two field visits are conducted for each student.

-What would make supervising students easier for you

Having a system that organizes reports would make supervision easier. It would also be better if the system allows evaluation without the need to download a large number of files.

-Would you prefer a system that organizes reports, tracks progress, and manages communication? Why?

Yes, because it makes tracking and management easier and more efficient.

Supervisor 2

-How are students assigned to you as a supervisor?

They are assigned based on specialization and department distribution.

-Is the assignment based on students' preference?

Rarely, most assignments are based on availability and distribution.

-Are you assigned to supervise more than one student?

Yes, I supervise several students at the same time.

-What information do you receive about each student at the beginning of the training?

Basic information only such as name student section number, and student ID number.

-How do you currently communicate with your students?

Through email or university systems.

-How often do you communicate with students during the training period?

When needed, not on a regular weekly basis.

-Is there communication between you and the university supervisor?

Sometimes problems arise, so we need to communicate with the supervisors at companies.

-If yes, how does this communication take place?

Through email or occasional meetings such as Zoom or Microsoft Teams.

-If no, would you prefer having such communication?

Yes, I would prefer more consistent communication.

-How do students submit their weekly reports?

Through Blackboard, and in some cases via email if needed.

-In what format are the reports submitted?

Usually in PDF format or word.

-What information should each report include?

A description of the weekly tasks, what the student learned, and any challenges faced.

-How do you review and evaluate the reports?

I review the submitted reports and provide feedback based on the content. I also make the evaluation based on the student's skills and the work completed. The assessment is done through the submitted files on Blackboard.

-How do you track student progress over time?

Indirectly through the submitted reports, especially toward the end of the training.

-Do you keep records of student reports and performance? If yes, how?

Yes, I keep records by saving reports and evaluation notes electronically.

-What challenges do you face when managing multiple students?

It is difficult to closely monitor all students, especially with a large number of reports to review.

-Is it difficult to track all student submissions and feedback?

Yes, it can be difficult to keep track of all submissions and feedback.

-How do you perform the final evaluation of students?

Based on overall performance, submitted reports, and any available feedback.

-What criteria do you use for evaluation?

The evaluation is based on commitment, the quality of submitted reports, and overall performance, in addition to any available feedback from the training organization.

-What is the biggest difficulty you face during the supervision process?

The biggest issue is the lack of easy coordination with students and company supervisors, which makes follow-up and scheduling more difficult.

-What are the main stages students go through during the training period?

The stages include the start of training, continuous follow-up through reports, and final evaluation.

-How many field visits are usually conducted during the training?

Usually one visit, but it is often difficult to schedule due to time constraints and coordination issues.

-What would make supervising students easier for you?

A unified system that organizes reports and helps in scheduling field visits would make supervision easier.

-Would you prefer a system that organizes reports, tracks progress, and manages communication? Why?

Yes, because it would save time and improve tracking. It would also help solve the issue of not having direct communication with the company supervisor, which currently makes coordination more difficult.

Appendix D:

Company Interview:

Questions for the Administration Responsible for Student Admission Employee A, Employee A, personal communication, May 11 2026.

Company A Meeting Confirmation

Employee B, personal communication, May 11 2026.

Company B Meeting Confirmation

Company Interview questions:

Questions for the Administration Responsible for Student Admission

Employee A, personal communication, May 11 2026.
Company A Meeting Confirmation
Employee B, personal communication, May 11 2026.
Company B Meeting Confirmation

1. Are you required to follow a specific training schedule (fixed dates), or does it depend on your internal workflow?

Company A - Yes, the training durations are specified for each opportunity

Company B – Yes, the training durations are specified for each opportunity or depending on the university specification

2. What are the steps a student goes through from application to acceptance?

Company A – First, they go through their qualifications then they set up an interview with students

Company B – They set up an interview with students then review their skills during the interview

3. After the acceptance process is completed, is communication limited to the student and supervisor, or does the administration continue to communicate with the student? If yes, how is communication handled during the training period?

Company A—Most of the time it's just the student and supervisor but if there are any issues it gets escalated to the managers. It's handled through emails or phone numbers.

Company B – Most of the time it's just the student and supervisor but if there are any issues it gets escalated to the managers. Through emails or phone numbers there is no centralized platform for communication

4. How do you store and manage students' and applicants' data?

Company A— sometimes manually, or through their system

Company B – through their system

5. Do you experience high pressure due to the number of applications? How are applications usually filtered and responded to?

Company A— yes, sometimes there are more students than we can accept for training, by going through each application sent to them.

Company B – yes, by going through each application sent to them.

6. Which stage of the process takes the most time or causes the most issues?

Company A— Monitoring students causes a lot of issues and accepting them till they start training takes a lot of time

Company B – Monitoring students' progress

7. Are there specific criteria or requirements for accepting students?

Company A— Their major and depending on the role the technical skills as well as soft skills

Company B – Their major, certifications and soft skills

8. How do you assign students to departments or supervisors?

Company A— by checking their requirements, and by checking if the supervisors have reached the limit for students or not.

Company B – by their major, and any supervisor that's available.

9. Have you ever faced issues where a student was not suitable for the training?

Company A— yes many

Company B – yes

10. Is there a standardized evaluation or reporting form? And is the training plan fixed?

Company A— No there are none

Company B – Yes, only a survey we are provided with at the end of the training

11. Do you face issues with student delays or lack of commitment? How do you handle such cases?

Company A, B— yes , usually it gets escalated to the university supervisor

Second: Questions Focused on Daily Student Follow-up (Supervisor)

12. How is a student assigned under your supervision?

Company A— by the HR management

Company B – by the HR management

13. How do you track the student's progress during the training period?

Company A— by assigning tasks and checking if they have completed them successfully

Company B – by assigning tasks and allowing them to give presentations based on them to check their understanding and work completion.

14. Are there specific tasks that the student is required to complete? How do you evaluate their performance in completing them?

Company A, B— yes, usually by if they were completed correctly.

15. Are there reports that need to be reviewed by the supervisor before submission to the university? If yes, how do you receive these reports or tasks from the student?

Company A, B — there's usually no direct access to the reports. The student submits on their own, but it should be evaluated. And occasionally students show us the reports to evaluate them.

16. Is it possible for a supervisor to be assigned to more than one student? If yes, do you face difficulty in managing multiple trainees?

Company A, B— yes, tracking their progress in tasks and monitoring them becomes difficult to manage. it would be preferred to have a set limit.

17. What is the most common issue you face with trainees?

Company A, B—their attendance.

18. How do you communicate with students?

Company A, B— through email and phone numbers.

19. What are the most important factors you focus on during evaluation?

Company A, B— Task completion and soft skills.

20. Is there an official evaluation form?

Companies A, B — No

21. How do you communicate with the university supervisor?

Company A, B— Through email, phone numbers, whatsapp and when they visit

22. Would you prefer having a system that centralizes communication and reports in one place?

Company A, B— yes

23. Do you need access to the student's data and skills before the training begins?

Company A, B— most of the time no but it would be preferred

24. What features would you like to have to make supervision easier?

Company A, B— a platform where we can communicate with both students and supervisors at the same time and allows us to evaluate and monitor student reports and an online survey that we can fill out to give students the final evaluation on.